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1 Data Structure

1.1 pbds

```
#include <ext/rope>
using namespace __gnu_cxx;
rope<char> rp;
rope<char> rp("string literal");
rope<char> rp(st.c_str()); // rp(st)는 안 됨
rp.push_back(ch);
rp.insert(idx, string or rope or char);
rp.erase(pos, cnt); // pos부터 cnt개 삭제
rp.substr(pos, cnt);
rp.substr(pos);  // rp[idx] 문자 하나. string에서
st.substr(idx)가 st[idx:]을 의미했던 것과는 다름에 주의
rope<char> rp = rp1 + rp2;
cout << rp;
for (auto it = rp.begin(); it != rp.end(); it++) //
rp[idx]접근은 O(logN)이므로 순회 시 iterator사용 // it++는
Amortized O(1)

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <typename T> using ordered_set = tree<T, null_type,
less<T>, rb_tree_tag,
tree_order_statistics_node_update>;
rb_tree_tag, tree_order_statistics_node_update>;
// find_by_order(k) : k(0-based)번째 값의 it 반환
// order_of_key(x) : x보다 작은 원소 개수 반환
ordered_set os;
os.insert(key);
os.erase(os.find_by_order(k));
```

```
os.erase(os.find_by_order(os.order_of_key(key))); // key 삭제 1
int order = os.order_of_key(key); // key 삭제 2 (들어있는값인지 확인후삭제)
if (order < os.size()) {
  auto it = os.find_by_order(os.order_of_key(key));
  if (*it == key) os.erase();
}
int order = os.order_of_key(key); // 검색
if (order < os.size() && *os.find_by_order(order) == val) ;
```

1.2 XOR Basis

```
vector<T> basis;
auto insert = [&](T x) {
  for (auto e : basis) x = min(x, x ^ e);
  if (x) basis.push_back(x);
};
T mxSum = 0;
auto b = basis; // insert를 계속해야하는 경우 basis의 순서는 바뀌면 안됨
sort(b.rbegin(), b.rend());
for (auto e : b) mxSum = max(mxSum, mxSum ^ e);
```

1.3 DSU Potential

```
template <typename T>
struct DSU {
  vector<int> p;
  vector<T> w;
  DSU(int n) : p(n + 1), w(n + 1) {
    iota(p.begin(), p.end(), 0);
  }
  int find(int a) {
    if (p[a] == a) return a;
    int ra = find(p[a]); // w[p[a]] 계산
    w[a] += w[p[a]];
    return p[a] = ra;
  }
  void merge(int a, int b, T val) { // b = a + val
    int ra = find(a);
    int rb = find(b);
    w[rb] = w[a] + val - w[b];
    p[rb] = ra;
  }
  T getDiff(int a, int b) {
    int ra = find(a);
    int rb = find(b);
    assert(ra == rb);
    return w[b] - w[a];
  }
};
```

1.4 Fenwick

```
template <typename T, int D>
struct FenwickTree {
  vector<FenwickTree<T, D - 1>> tree;
  template <typename... Args>
  FenwickTree(int n, Args... args) : tree(n + 1,
  FenwickTree<T, D - 1>(args...)) {}
  template <typename... Args>
  void upd(int pos, Args... args) {
    for (; pos < tree.size(); pos += (pos & -pos))
    tree[pos].upd(args...);
  }
  template <typename... Args>
  T sum(int r, Args... args) const {
    T res = 0;
    for (; r; r -= (r & -r)) res += tree[r].qry(args...);
    return res;
  }
  template <typename... Args>
  T qry(int l, int r, Args... args) const {
    return sum(r, args...) - sum(l - 1, args...);
  }
};
template <typename T>
struct FenwickTree<T, 0> {
  T val = 0;
  void upd(T v) { val += v; }
  T qry() const { return val; }
};
// ex) 3D fenwick
// FenwickTree<int, 3> fw(n, m, k);
// fw.upd(x, y, z, add); // O(logN logM logK)
// fw.qry(x1, x2, y1, y2, z1, z2); // O(logN logM logK)
```

1.5 Segtree Walking

```
template <typename F>
int rightMost(int l, F f) const { // f([l:r])=true인 최대 r // 만족하는 r 없다면 l-1리턴
  int node = l | sz;
  T acc = e();
  node >>= __builtin_ctz(node);
  while (f(op(acc, tree[node]))) {
    acc = op(acc, tree[node++]);
    if (node & (node - 1)) node >>= __builtin_ctz(node);
    else return n - 1;
  }
  while (node < sz) {
    node <<= 1;
    if (f(op(acc, tree[node]))) acc = op(acc, tree[node++]);
  }
  return (node ^ sz) - 1;
}
template <typename F>
int leftMost(int r, F f) const { // f([l:r])=true인 최소 l // 만족하는 l 없다면 r+1리턴
  int node = r | sz;
  T acc = e();
  node = max(1, node >> __builtin_ctz(~node));
  while (f(op(tree[node], acc))) {
    acc = op(tree[node--], acc);
    if ((node + 1) & node) node = max(1, node >>
    __builtin_ctz(~node));
    else return 0;
  }
  while (node < sz) {
    node = node << 1 | 1;
    if (f(op(tree[node], acc))) acc = op(tree[node--], acc);
  }
  return (node ^ sz) + 1;
}
int kthSmallest(int l, T k) const { // qry([l, r]) >= k인 최소 r
  if (qry(l, n - 1) < k) return -1;
  return rightMost(l, [&](T sum) { return sum < k; }) + 1;
}
int kthLargest(int r, T k) const { // qry([l, r]) >= k인 최대 l
  if (qry(0, r) < k) return -1;
  return leftMost(r, [&](T sum) { return sum < k; }) - 1;
}
```

1.6 Sweeping Segtree

```
int sz = 1;
while (sz < n) sz <<= 1;
vector<int> tree(sz << 1), cnt(sz << 1);
auto upd = [&](auto &&upd, int node, int s, int e, int l, int r, int add) -> void {
  if (l > e || s > r) return;
  if (l <= s && e <= r) cnt[node] += add;
  else {
    int m = s + e >> 1;
    upd(upd, node << 1, s, m, l, r, add);
    upd(upd, node << 1 | 1, m + 1, e, l, r, add);
  }
  if (cnt[node]) tree[node] = e - s + 1;
  else tree[node] = s != e ? tree[node << 1] + tree[node << 1 | 1] : 0;
};
auto qry = [&]() { return tree[1]; };
```

1.7 PST

```
struct Node {
  int l, r;
  ll val;
  Node() { l = r = val = 0; }
};
vector<Node> tree;
auto newNode = [&]() -> int {
  tree.emplace_back();
  return tree.size() - 1;
};
auto upd = [&](auto &&upd, int old, int s, int e, int i, ll add) {
  int node = newNode();
  tree[node] = tree[old];
  if (s == e) {
    tree[node].val += add;
```

```
    return node;
  }
  int m = s + e >> 1;
  if (i <= m) tree[node].l = upd(upd, tree[old].l, s, m, i,
  add);
  else tree[node].r = upd(upd, tree[old].r, m + 1, e, i, add);
  tree[node].val = tree[tree[node].l].val +
  tree[tree[node].r].val;
  return node;
};
auto qry = [&](auto &qry, int node, int s, int e, int l, int
r) {
  if (l <= s && e <= r) return tree[node].val;
  if (l > e || s > r) return 0LL;
  int m = s + e >> 1;
  return qry(qry, tree[node].l, s, m, l, r) + qry(qry,
  tree[node].r, m + 1, e, l, r);
};
vector<int> roots = {newNode()}; // version 0
```

1.8 treap

```
constexpr ll MOD = 998'244'353;
struct Node {
  int l = -1, r = -1;
  ll sum, lm = 1, la = 0, val;
  int pr, sz = 1;
  bool flip = false;
  inline static mt19937 rng;
  Node(int val) : sum(val), val(val), pr(rng()) {}
  void propagate();
  void recalc();
};
vector<Node> tree;
int cnt(int node) { return ~node ? tree[node].sz : 0; }
void Node::propagate() {
  if (flip) {
    swap(l, r);
    if (~l) tree[l].flip ^= 1;
    if (~r) tree[r].flip ^= 1;
    flip = 0;
  }
  if (lm != 1 || la) {
    val = (val * lm + la) % MOD;
    sum = (sum * lm + la * sz) % MOD;
    for (auto nxt : {l, r}) if (~nxt) {
      tree[nxt].lm = tree[nxt].lm * lm % MOD;
      tree[nxt].la = (tree[nxt].la * lm + la) % MOD;
    }
    lm = 1;
    la = 0;
  }
}
void Node::recalc() {
  sz = 1;
  sum = val;
  if (~l) {
    tree[l].propagate();
    sz += tree[l].sz;
    sum += tree[l].sum;
    if (sum >= MOD) sum -= MOD;
  }
  if (~r) {
    tree[r].propagate();
    sz += tree[r].sz;
    sum += tree[r].sum;
    if (sum >= MOD) sum -= MOD;
  }
}
pair<int, int> split(int node, int k) {
  if (!~node) return {-1, -1};
  tree[node].propagate();
  if (cnt(tree[node].l) >= k) {
    auto [a, b] = split(tree[node].l, k);
    tree[node].l = b;
    tree[node].recalc();
    return {a, node};
  }
  auto [a, b] = split(tree[node].r, k - cnt(tree[node].l) -
  1);
  tree[node].r = a;
  tree[node].recalc();
  return {node, b};
}
int merge(int a, int b) {
  if (!~a) return b;
  if (!~b) return a;
  tree[a].propagate();
  tree[b].propagate();
  if (tree[a].pr > tree[b].pr) {
    tree[a].r = merge(tree[a].r, b);
    tree[a].recalc();
    return a;
  }
  tree[b].l = merge(a, tree[b].l);
  tree[b].recalc();
  return b;
}
void insert(int &root, int idx, int node) {
  auto [a, b] = split(root, idx);
  root = merge(merge(a, node), b);
}
void erase(int &root, int idx) {
  auto [a, bc] = split(root, idx);
  auto [b, c] = split(bc, 1);
  root = merge(a, c);
}

template <typename F>
void f(int &root, int l, int r, F f) {
  auto [a, bc] = split(root, l);
  auto [b, c] = split(bc, r - l + 1);
  assert(~b);
  f(b); // split후라서 b는 propagate된 상태
  root = merge(merge(a, b), c);
}

// reverse(root, l, r)
f(root, l, r, [&](int node) {
  tree[node].flip ^= 1;
});
// upd(root, l, r, mul, add)
f(root, l, r, [&](int node) {
  tree[node].lm = mul;
  tree[node].la = add;
});
// qry(root, l, r)
f(root, l, r, [&](int node) {
  cout << tree[node].sum << "\n";
});
```

1.9 Cartesian Tree

```
auto getCartesian(const vector<int> &v) { // min // 0-based
  int n = v.size();
  vector<pair<int, int>> chd(n, {-1, -1});
  vector<int> s;
  for (int i = 0; i < n; i++) {
    while (!s.empty() && v[s.back()] > v[i]) {
      chd[i].first = s.back();
      s.pop_back();
    }
    if (!s.empty()) chd[s.back()].second = i;
    s.push_back(i);
  }
  return pair{s[0], chd}; // root, chd
}
// i의 왼쪽 직계 자식에서 오른쪽 방향으로만 이동하는 체인 위의
j에 대해 right_lt[j]=i
// i의 오른쪽 직계 자식에서 왼쪽 방향으로만 이동하는 체인 위의
j에 대해 left_le[j]=i
```

1.10 Line Container

```
// upper=false일 때 테스트안해봤음
struct Line { //
https://github.com/kth-competitive-programming/kactl/blob/main/conten
  mutable ll k, m, p;
  bool operator<(const Line& o) const { return k < o.k; }
  bool operator<(ll x) const { return p < x; }
};
template <bool upper>
struct LineContainer : multiset<Line, less<>> {
  // (for doubles, use inf = 1/.0, div(a,b) = a/b)
  static const ll inf = LLONG_MAX;
  ll div(ll a, ll b) { // floored division
    return a / b - ((a ^ b) < 0 && a % b);
  }
  bool isect(iterator x, iterator y) {
```

```
    if (y == end()) return x->p = inf, 0;
    if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
    else x->p = div(y->m - x->m, x->k - y->k);
    return x->p >= y->p;
  }
  void add(ll k, ll m) {
    if constexpr (!upper) k = -k, m = -m;
    auto z = insert({k, m, 0}), y = z++, x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
    while ((y = x) != begin() && (--x)->p >= y->p) isect(x,
    erase(y));
  }
  ll qry(ll x) {
    assert(!empty());
    auto l = *lower_bound(x);
    if constexpr (upper) return l.k * x + l.m;
    else return -(l.k * x + l.m);
  }
};
```

2 DP

2.1 knuth opt

```
const int INF = 2e9;
vector dp(k + 1, vector<int>(n + 1, INF));
for (int i = 0; i <= k; i++) dp[i][0] = 0;

auto getDP = [&](auto &&getDP, int i, int s, int e, int optL,
int optR) -> void {
  if (s > e) return;
  int m = s + e >> 1;
  int opt = -1;
  for (int j = max(lowerLimit(m), optL); j <=
  min(upperLimit(m), optR); j++) {
    int val = dp[i - 1][j] + f(j + 1, m);
    if (dp[i][m] > val) {
      dp[i][m] = val;
      opt = j;
    }
  }
  assert(~opt);
  getDP(getDP, i, s, m - 1, optL, opt);
  getDP(getDP, i, m + 1, e, opt, optR);
};

for (int i = 1; i <= k; i++) getDP(getDP, i, 1, n, 0, n - 1);
// O(KN logN)
cout << dp[k][n];

// dp[i][k] = min_{low(k)<=j<=up(k)}{ dp[i - 1][j] + C(j, k) }
// dp[i][k]에 대해 dp[i - 1][j] + C(j, k)가 최소가 되는 가장
작은 j를 opt(i, k)라 할 때 opt(i, k) <= opt(i, k + 1)이어야 함
// 사각부등식(a<=b<=c<=d일 때 f(a,c)+f(b,d)<=f(a,d)+f(b,c))
만족하면 opt(i, j) <= opt(i, j + 1)
```

2.2 DnC opt

```
const int INF = 2e9;
vector dp(k + 1, vector<int>(n + 1, INF));
for (int i = 0; i <= k; i++) dp[i][0] = 0;

auto getDP = [&](auto &&getDP, int i, int s, int e, int optL,
int optR) -> void {
  if (s > e) return;
  int m = s + e >> 1;
  int opt = -1;
  for (int j = max(lowerLimit(m), optL); j <=
  min(upperLimit(m), optR); j++) {
    int val = dp[i - 1][j] + f(j + 1, m);
    if (dp[i][m] > val) {
      dp[i][m] = val;
      opt = j;
    }
  }
  assert(~opt);
  getDP(getDP, i, s, m - 1, optL, opt);
  getDP(getDP, i, m + 1, e, opt, optR);
};

for (int i = 1; i <= k; i++) getDP(getDP, i, 1, n, 0, n - 1);
// O(KN logN)
cout << dp[k][n];

// dp[i][k] = min_{low(k)<=j<=up(k)}{ dp[i - 1][j] + C(j, k) }
// dp[i][k]에 대해 dp[i - 1][j] + C(j, k)가 최소가 되는 가장
작은 j를 opt(i, k)라 할 때 opt(i, k) <= opt(i, k + 1)이어야 함
// 사각부등식(a<=b<=c<=d일 때 f(a,c)+f(b,d)<=f(a,d)+f(b,c))
만족하면 opt(i, j) <= opt(i, j + 1)
```

2.3 hirschberg

```
vector<int> getDP(const string &a, const string &b) {
  int n = a.size(), m = b.size();
  vector<int> dp(m + 1);
  for (int i = 1; i <= n; i++) {
    vector<int> ndp(m + 1);
    for (int j = 1; j <= m; j++) {
      if (a[i - 1] == b[j - 1]) ndp[j] = dp[j - 1] + 1;
      else ndp[j] = max(ndp[j - 1], dp[j]);
    }
    swap(dp, ndp);
  }
  return dp;
}

string LCS(const string &a, const string &b) { // time O(NM),
space O(M)
  if (a.empty()) return "";
  if (a.size() == 1) return b.find(a) != string::npos ? a :
  "";
  int n = a.size(), m = b.size();
  int mid = n / 2;
  string l = a.substr(0, mid), r = a.substr(mid);
  auto dp1 = getDP(l, b);
  string rrev = r, brev = b;
  reverse(rrev.begin(), rrev.end());
  reverse(brev.begin(), brev.end());
  auto dp2 = getDP(rrev, brev);
  int mx = -1, mxIdx;
  for (int i = 0; i <= m; i++) {
    if (mx < dp1[i] + dp2[m - i]) {
      mx = dp1[i] + dp2[m - i];
      mxIdx = i;
    }
  }
  return LCS(l, b.substr(0, mxIdx)) + LCS(r, b.substr(mxIdx));
}
```

3 Geometry

3.1 Header

```
template <typename T>
struct Point {
  T x, y;
  Point() = default;
  Point(T x, T y) : x(x), y(y) {}
  template <typename U> Point(const Point<U> &other) :
  x(static_cast<T>(other.x)), y(static_cast<T>(other.y)) {}
  bool operator<(const Point &other) const { return tie(x, y)
  < tie(other.x, other.y); }
  bool operator<=(const Point &other) const { return tie(x, y)
  <= tie(other.x, other.y); }
  bool operator==(const Point &other) const { return tie(x, y)
  == tie(other.x, other.y); }
  Point operator-(const Point &other) const { return
  {x-other.x, y-other.y}; }
  Point operator+(const Point &p) const { return {x+p.x,
  y+p.y}; }
  Point operator*(T k) const { return {x*k, y*k}; }
  Point operator/(T k) const { return {x/k, y/k}; }
  friend T dot(Point a, Point b) { return a.x*b.x+a.y*b.y; }
  friend T cross(Point a, Point b) { return a.x*b.y-a.y*b.x; }
  friend int ccw(Point a, Point b, Point c) {
    T v = cross(b-a,c-a);
    return (v>0)-(v<0);
  }
  friend bool onPS(Point p, Point a, Point b) {
    return ccw(a, b, p) == 0 && dot(p - a, p - b) <= 0;
  }
};
// P: Point, L: Line, S: Segment, H: convexHull
auto footPL(Point<ld> p, Point<ld> a, Point<ld> b) {
  auto d = b - a;
  return a + d * (dot(p - a, d) / dot(d, d));
}
auto reflectPL(Point<ld> p, Point<ld> a, Point<ld> b) {
  return footPL(p, a, b) * 2 - p;
}
```

3.2 Fomular

```
Point<ld> rotate2d(Point<ld> p, ld theta) { // 반시계 theta
  ld c = cos(theta), s = sin(theta);
  return {c * p.x - s * p.y, s * p.x + c * p.y};
}
pll tangentMerge(ll a, ll b, ll c, ll d) { // tanT1 = a / b,
tanT2 = c / d, tan(T1 + T2) = ?
  return {
    (a * d % mod + b * c % mod) % mod,
    (b * d % mod - a * c % mod + mod) % mod
  }; // {분자, 분모}
}

template <typename T>
inline T dist2PP(const Point<T> &p1, const Point<T> &p2) {
  return (p2.x - p1.x) * (p2.x - p1.x) + (p2.y - p1.y) * (p2.y
  - p1.y);
}
template <typename T>
inline ld distPP(const Point<T> &p1, const Point<T> &p2) {
  return hypot<ld>(p2.x - p1.x, p2.y - p1.y);
}
template <typename T>
ld distPL(const Point<T> &p, const Point<T> &l1, const
Point<T> &l2) { // distance from P(point, p) to L(line, l1l2)
  assert(!(l1 == l2));
  return abs(cross(l1 - p, l2 - p)) / distPP(l1, l2);
}

template <typename T>
T area2G(const vector<Point<T>> &g) {
  if (g.size() <= 2) return 0;
  T res = 0;
  for (int i = 0, j = g.size() - 1; i < g.size(); j = i++) res
  += cross(g[j], g[i]);
  return abs(res);
}
ld heron(ld a, ld b, ld c) { // 헤론
  ld s = ld(0.5) * (a + b + c);
  return sqrt(s * (s - a) * (s - b) * (s - c));
}
ld brahmagupta(ld a, ld b, ld c, ld d) { // 브라마굽타,
cyclic이어야 함
  ld s = ld(0.5) * (a + b + c + d);
  return sqrt((s - a) * (s - b) * (s - c) * (s - d));
}
ld getSegmentCircleArea(ld r, ld len) { // 활꼴 넓이, r :
반지름, len : 활꼴 길이
  ld cosTheta = 1 - ld(len * len) / (2 * r * r);
  ld sinTheta = sqrt(1 - cosTheta * cosTheta); // > 0
  ld theta = acos(cosTheta);
  return ld(0.5) * r * r * (theta - sinTheta);
}
```

3.3 Clipping

```
auto clipHL(const vector<Point<ld>> &h, Point<ld> a, Point<ld>
b) { // a->b의 왼쪽만 남김
  vector<Point<ld>> res;
  for (int i = 0, j = int(h.size()) - 1; i < h.size(); j =
  i++) {
    auto p = h[j], q = h[i];
    ld s = cross(b - a, p - a), t = cross(b - a, q - a);
    if (s >= 0) res.push_back(p);
    if ((s < 0 && t > 0) || (s > 0 && t < 0)) res.push_back(p
    + (q - p) * (s / (s - t)));
  }
  return res;
}
vector<Point<ld>> interHH(vector<Point<ld>> h1, const
vector<Point<ld>> &h2) { // O(M(N+M))
  // h2는 반시계여야 됨
  if (h2.empty()) return {};
  for (int i = 0, j = h2.size() - 1; i < h2.size(); j = i++)
  h1 = clipHL(h1, h2[j], h2[i]);
  return h1;
}
```

3.4 Smallest Enclosing Circle

```
template <typename T>
Point<ld> circumcircle(const Point<T> &p1, const Point<T> &p2,
const Point<T> &p3) {
  Point<T> b = p2 - p1, c = p3 - p1;
  T dist12 = b.x * b.x + b.y * b.y;
  T dist23 = c.x * c.x + c.y * c.y;
  T d = 2 * cross(b, c);
  return Point<ld>(p1.x + ld(dist12 * c.y - dist23 * b.y) / d,
  p1.y + ld(dist23 * b.x- dist12 * c.x) / d);
}
template <typename T>
pair<Point<ld>, ld>
getSmallestEnclosingCircle(vector<Point<T>> ps) { // 평균 O(N)
  mt19937
  rng(chrono::system_clock::now().time_since_epoch().count());
  shuffle(ps.begin(), ps.end(), rng);
  Point<ld> res;
  ld r = -1;
  for (int i = 0; i < ps.size(); i++) if (distPP(res, ps[i]) >
  r) {
    res = ps[i], r = 0;
    for (int j = 0; j < i; j++) if (distPP(res, ps[j]) > r) {
      res = Point<ld>(ld(ps[i].x + ps[j].x) / 2, ld(ps[i].y +
      ps[j].y) / 2), r = distPP(res, ps[i]);
      for(int k = 0; k < j; k++) if (distPP(res, ps[k]) > r)
      res = circumcircle(ps[i], ps[j], ps[k]), r = distPP(res,
      ps[k]);
    }
  }
  return {res, r};
}
```

3.5 Segment Intersection

```
template <typename T>
bool checkSS(const Point<T> &p1, const Point<T> &p2, const
Point<T> &p3, const Point<T> &p4) {
  int a = ccw(p1, p2, p3) * ccw(p1, p2, p4);
  int b = ccw(p3, p4, p1) * ccw(p3, p4, p2);
  if (a > 0 || b > 0) return false;
  if (a < 0 || b < 0) return true;
  return min(p1, p2) <= max(p3, p4) && min(p3, p4) <= max(p1,
  p2);
}
```

3.6 Convex Hull

```
template <typename T>
auto getConvexHull(vector<Point<T>> ps) { // O(NlogN)
  sort(ps.begin(), ps.end());
  ps.erase(unique(ps.begin(), ps.end()), ps.end());
  if (ps.size() <= 1) return ps;
  vector<Point<T>> h;
  for (auto x : ps) {
    while (h.size() >= 2 && ccw(h[h.size() - 2], h.back(), x)
    <= 0) h.pop_back();
    h.push_back(x);
  }
  int lower = h.size();
  for (int i = ps.size() - 2; i >= 0; i--) {
    while (h.size() > lower && ccw(h[h.size() - 2], h.back(),
    ps[i]) <= 0) h.pop_back();
    h.push_back(ps[i]);
  }
  h.pop_back();
  return h;
}
```

3.7 Rotating Calipers

```
template <typename T>
T getDiameterSquare(const vector<Point<T>> &h) { // O(NlogN)
  // h 반시계여야함
  if (h.size() <= 1) return 0;
  if (h.size() == 2) return dist2PP(h[0], h[1]);
  T res = 0;
  int n = h.size(), a = 0, b = 1;
  while (a < n && b < 2 * n) {
    res = max(res, dist2PP(h[a % n], h[b % n]));
    if (cross(h[(a + 1) % n] - h[a % n], h[(b + 1) % n] - h[b
    % n]) >= 0) ++b;
    else ++a;
  }
  return res;
}
template <typename T>
inline T f(const Point<T> &a, const Point<T> &b, const
Point<T> &c) {
  return cross(b - a, c - a);
}
template <typename T>
```

```
T largestTriangleDouble(const vector<Point<T>> &h) { // O(N^2)
// 볼록껍질에 포함되는 가장 큰 삼각형의 넓이
  assert(h.size() >= 3);
  T res = 0;
  for (int i = 0; i + 2 < h.size(); i++) {
    for (int j = i + 1, k = i + 2; j < h.size(); j++) {
      while (k + 1 < h.size() && f(h[i], h[j], h[k]) < f(h[i],
      h[j], h[k + 1])) ++k;
      res = max(res, f(h[i], h[j], h[k]));
    }
  }
  return res;
}
```

3.8 DnC

```
constexpr ld eps = 1e-5;
ld dnc(vector<pair<ll, ll>> ps) { // O(N log^2N)
  assert(ps.size() > 1);
  sort(ps.begin(), ps.end(), [&](auto &a, auto &b) { return
  a.first < b.first; });
  auto f = [&](auto &&f, int s, int e) -> ld {
    if (s + 1 == e) return dist(ps[s], ps[e]);
    if (s + 2 == e) return min({
      dist(ps[s], ps[s + 1]),
      dist(ps[s + 1], ps[e]),
      dist(ps[e], ps[s])
    });
    int m = s + e >> 1;
    ld res = min(f(f, s, m), f(f, m + 1, e));
    vector<pair<ll, ll>> mids;
    for (int i = s; i <= e; i++) {
      if (abs(ps[m].first - ps[i].first) < res)
      mids.push_back(ps[i]);
    }
    // TLE나면 merge사용
    sort(mids.begin(), mids.end(), [&](auto &a, auto &b) {
    return a.second < b.second; });
    for (int i = 0; i < mids.size() - 1; i++) {
      for (int j = i + 1; j < mids.size(); j++) {
        if (mids[j].second - mids[i].second >= res) break;
        res = min(res, dist(mids[i], mids[j]));
      }
    }
    return res;
  };
  return f(f, 0, ps.size() - 1);
}
```

3.9 PIP

```
template <typename T>
int PIP(const Point<T> &p, const vector<Point<T>> &g) { // -1
: 내부, 0 : 경계, 1 : 외부
  int n = g.size(), in = 0;
  assert(n >= 3);
  for (int i = 0, j = n - 1; i < n; j = i++) {
    if (onPS(p, g[i], g[j])) return 0;
    if ((g[i].y > p.y) != (g[j].y > p.y)) in ^= (ccw(g[i],
    g[j], p) < 0) == (g[i].y < g[j].y);
  }
  return 1 - 2 * in;
}
```

3.10 PICP

```
template <typename T>
int PICP(const Point<T> &p, const vector<Point<T>> &h) { // -1
: 내부, 0 : 경계, 1 : 외부
  // h : ccw, no collinear
  assert(h.size() >= 3);
  if (int c = ccw(h[0], h[1], p); c <= 0) return c < 0 ||
  dot(p - h[0], p - h[1]) > 0;
  if (int c = ccw(h[0], h.back(), p); c >= 0) return c > 0 ||
  dot(p - h[0], p - h.back()) > 0;
  auto it = upper_bound(h.begin() + 1, h.end(), p, [&](const
  auto &a, const auto &b) { return ccw(h[0], a, b) > 0; });
  return -ccw(*prev(it), *it, p);
}
```

3.11 Tangent

```
template <typename T, typename F>
int extremeVertex(const vector<Point<T>> &h, F direction) {
  // 공선점 없어야됨, h반시계
  int n = h.size();
  assert(n >= 3);
  auto cmp = [&](int i, int j) {
    T v = cross(direction(h[j]), h[j] - h[i]);
    return (v > 0) - (v < 0);
  };
  auto slope = [&](int i) {
    return cmp((i + 1) % n, i);
  };
  auto check = [&](int i) {
    return slope(i) >= 0 && cmp(i, (i + n - 1) % n) < 0;
  };
  int l = 0, r = check(0) ? 1 : n;
  int s = slope(0);
  while (l + 1 < r) {
    int m = (l + r) / 2, t = slope(m);
    if (check(m)) return m;
    if (s != t ? s < t : s == cmp(l, m)) r = m;
    else l = m, s = t;
  }
  return l;
}
// 다각형의 모든 꼭짓점이 "p를 지나며 direction(p)방향의 직선"
오른쪽에 놓이게 되는 p의 인덱스 리턴
// 즉, cross(direction(p), h[i]-p)<=0
// 접선이 다각형의 변과 접하는 경우, 그 변에서 ccw순서상
첫번째점이 반환됨

auto tangent = [&](Point<T> p, const vector<Point<T>> &h) {
  // p가 h에 대해 strictly outside여야됨
  int a = extremeVertex(h, [&](Point<T> q) { return q - p; });
  int b = extremeVertex(h, [&](Point<T> q) { return p - q; });
  return pair{a, b};
};
int i = extremeVertex(h, [&](Point<T> q) { return dir; });
```

3.12 Bulldozer

```
template <typename T>
struct SlopeWithIdx {
  int i, j; // idx
  T dx, dy; // 무조건 dx >= 0 되게 저장
  SlopeWithIdx(int i, int j, Point<T> dir) : i(i), j(j),
  dx(dir.x), dy(dir.y) {}
  bool operator<(const SlopeWithIdx &other) const { return
  tuple<T, int, int>{dy * other.dx, i, j} < tuple<T, int,
  int>{other.dy * dx, other.i, other.j}; }
  bool operator==(const SlopeWithIdx &other) const { return dy
  * other.dx == other.dy * dx; }
};
template <typename T>
void bulldozer(vector<Point<T>> &ps) { // O(N^2 logN)
  int n = ps.size();
  vector<int> pos(n);
  iota(pos.begin(), pos.end(), 0);
  sort(ps.begin(), ps.end());
  vector<SlopeWithIdx<T>> slopes;
  slopes.reserve(n * (n - 1) / 2);
  for (int i = 0; i < n; i++) for (int j = i + 1; j < n; j++)
  slopes.emplace_back(i, j, ps[j] - ps[i]);
  sort(slopes.begin(), slopes.end());
  for (int i = 0, j; i < slopes.size(); i = j) {
    for (j = i; j < slopes.size() && slopes[i] == slopes[j];
    j++) {
      int &idx1 = pos[slopes[j].i], &idx2 = pos[slopes[j].j];
      // 인접한 원소끼리 순서를 바꿔며 모든 정렬을 순회하게
      됨, 즉 abs(idx1-idx2)==1임
      swap(idx1, idx2);
      swap(ps[idx1], ps[idx2]);

      // idx1이랑 idx2는 pos[]의 alias이므로 함부로 값을
      수정하면 안됨
      // 예를 들어 if (idx1 > idx2) swap(idx1, idx2) 같은 거
      하면 안됨
      // 그냥 안전하게 i1, i2 선언해서 사용하는 거 추천
      int i1 = min(idx1, idx2);
      int i2 = max(idx1, idx2);
    }
    // slopes[i]의 기울기를 기준으로 정렬되어있음
    // 코드
  }

  // // 점들 뒤집는 nC2개의 모든 과정을 확인
  // for (int i = 0; i < slopes.size(); i++) { // 이 경우 for
  (auto &slope : slopes)로 해도 무방
  //     int &idx1 = pos[slopes[i].i], &idx2 =
  pos[slopes[i].j];
```

```
  //     swap(idx1, idx2);
  //     swap(ps[idx1], ps[idx2]);
  //     // 코드
  // }
}
```

3.13 HPI

```
const ld eps = 1e-9; // epsilon

template <bool isCCW>
struct HalfPlane {
  Point<ld> start, dir; // isCCW=tue일 때 반평면은 dir을
  반시계 90도 회전한 방향(dir 왼쪽)에 위치
  template <typename T>
  HalfPlane(const Point<T> &s, const Point<T> &e) : start(s),
  dir(e - s) {
    ld len = hypot(dir.x, dir.y);
    dir.x /= len;
    dir.y /= len;
  }
  template <typename T>
  HalfPlane(T a, T b, T c) { // ax + by + c >= 0
    assert(a != 0 || b != 0);
    start = (a != 0 ? Point<ld>(-ld(c) / a, 0) : Point<ld>(0,
    -ld(c) / b));
    // ax + by + c = 0이면 a(x+a) + b(y+b) + c =
    ax+by+c+a^2+b^2>=0 이므로 (a, b)쪽이 반평면 방향이 되어야
    함
    if constexpr (isCCW) dir = {b, -a};
    else dir = {-b, a};
    ld len = hypot(dir.x, dir.y);
    dir.x /= len;
    dir.y /= len;
  }
  template <typename T>
  static HalfPlane<isCCW> getBisector(const Point<T> &a, const
  Point<T> &b) { // 선분 ab의 수직이등분선이면서 dir이 a를
  가리키는 반평면 생성자
    ld sx = ld(a.x + b.x) / 2;
    ld sy = ld(a.y + b.y) / 2;
    if constexpr (isCCW) return HalfPlane<isCCW>(Point<ld>(sx,
    sy), Point<ld>(sx + (a.y - b.y), sy + (b.x - a.x)));
    else return HalfPlane<isCCW>(Point<ld>(sx, sy),
    Point<ld>(sx - (a.y - b.y), sy - (b.x - a.x)));
  }
  void move(ld d) {
    if constexpr (isCCW) { // dir 반시계 90도 법선 벡터 :
    (-dir.y, dir.x)
      start.x -= d * dir.y;
      start.y += d * dir.x;
    }
    else { // dir 시계 90도 법선 벡터 : (dir.y, -dir.x)
      start.x += d * dir.y;
      start.y -= d * dir.x;
    }
  }
  friend bool isParallel(const HalfPlane &a, const HalfPlane
  &b) { return abs(cross(a.dir, b.dir)) < eps; }
  friend bool isOpposite(const HalfPlane &a, const HalfPlane
  &b) { return (a.dir.x * b.dir.x + a.dir.y * b.dir.y) < -eps;
  }
  Point<ld> getPoint(ld k) const {
    return {start.x + k * dir.x, start.y + k * dir.y};
  }
  friend Point<ld> intersectionHH(const HalfPlane &a, const
  HalfPlane &b) { // assert (!isParallel(a, b))일 때만 호출 됨
    ld k = cross(b.start - a.start, a.dir) / cross(a.dir,
    b.dir);
    return b.getPoint(k);
  }
  bool include(const Point<ld> &point) { // point가 반평면에
  포함되는지
    if constexpr (isCCW) return cross(dir, point - start) >
    eps;
    else return cross(dir, point - start) < -eps;
  }
  bool include(const HalfPlane &a, const HalfPlane &b) { // a,
  b의 교점이 반평면에 포함되는지
    return include(intersectionHH(a, b));
  }
};
template <bool isCCW>
vector<Point<ld>> HPI(vector<HalfPlane<isCCW>> planes, ld d=0,
bool sortedByAngle=false) { // 반평면들을 각각 거리 d만큼
움직였을 때 교집합 // O(N logN)
  if (!sortedByAngle) { // 이미 선분들이 정렬되어있다면(ex)
  볼록다각형의 변들을 순서대로 넣은 경우) sortedByAngle=true
  사용
    auto checkQuadrant = [](const HalfPlane<isCCW> &hp) ->
    bool {
      return hp.dir.y < 0 || (hp.dir.y == 0 && hp.dir.x < 0);
      // PI <= atan2(dir) < 2 * PI
    };
    sort(planes.begin(), planes.end(), [&](const
    HalfPlane<isCCW> &a, const HalfPlane<isCCW> &b) {
      bool aq = checkQuadrant(a), bq = checkQuadrant(b);
      if constexpr (isCCW) return aq != bq ? aq < bq :
      cross(a.dir, b.dir) > 0;
      else return aq != bq ? aq < bq : cross(a.dir, b.dir) <
      0;
    });
  }
  vector<int> idxes; idxes.reserve(planes.size());
  for (int i = 0, j; i < planes.size(); i = j) {
    int cur = i;
    j = i + 1;
    while (j < planes.size() && isParallel(planes[i],
    planes[j])) {
      assert(!isOpposite(planes[i], planes[j]) && "교집합이
      다각형으로 제한되지 않고 무한히 넓음");
      if (planes[cur].include(planes[j].start)) cur = j;
      ++j;
    }
    idxes.push_back(cur);
  }
  assert(idxes.size() >= 3 && "교집합이 다각형으로 제한되지
  않고 무한히 넓음");
  // assert(!isParallel(planes[0], planes.back()) && "교집합이
  다각형으로 제한되지 않고 무한히 넓음");
  assert(!isParallel(planes[idxes[0]], planes[idxes.back()])
  && "교집합이 다각형으로 제한되지 않고 무한히 넓음");
  if (d) for (auto &plane : planes) plane.move(d);
  deque<HalfPlane<isCCW>> dq;
  for (auto idx : idxes) {
    auto &plane = planes[idx];
    while (dq.size() >= 2 && !plane.include(dq[dq.size() - 2],
    dq[dq.size() - 1])) dq.pop_back();
    while (dq.size() >= 2 && !plane.include(dq[0], dq[1]))
    dq.pop_front();
    if (dq.size() < 2 || dq[0].include(dq.back(), plane))
    dq.push_back(plane);
  }
  vector<Point<ld>> res;
  if (dq.size() < 3) return res; // 교집합의 크기가 점 하나에
  해당하는 경우 실수오차로 인해 교집합이 없는 것으로 계산되는
  경우 생김. 이 경우 distance-=1e-6로 살짝 옮겨주면 잘 계산됨
  for (int i = 0, j = dq.size() - 1; i < dq.size(); j = i++) {
    assert(abs(cross(dq[i].dir, dq[j].dir)) > eps); // TODO 이
    줄 그냥 지울까?
    res.push_back(intersectionHH(dq[i], dq[j]));
  }
  assert(dq[1].include(res[0]) && "교집합이 다각형으로
  제한되지 않고 무한히 넓음");
  return res; // 리턴값은 볼록 껍질 형태임, isCCW에 따라
  볼록껍질 정렬방향 다름(isCCW=true면 반시계방향정렬,
  isCCW=false면 시계방향정렬)
}

template <typename point_t, typename border_t>
vector<vector<Point<ld>>> getVoronoiDiagram(const
vector<Point<point_t>> &points, const vector<Point<border_t>>
&border, int dir=0) { // border은 시계/반시계 중 한 방향으로
정렬되있어야 함 // O(N^2 logN) // 중복점있으면안됨
  int n = border.size();
  if (!dir) { // dir은 border에서 점들이 주어진 방향(반시계 :
  1, 시계 : -1, 모르는 경우(default) : 0)
    int i = 2;
    while (!dir && i < n) dir = ccw(border[0], border[1],
    border[i++]);
  }
  assert(dir != 0); // dir = 0이면 모든 점이 일직선 위에 존재
  n = points.size();
  vector<vector<Point<ld>>> res(n);
```

```
  for (int i = 0; i < n; i++) {
    auto [x1, y1] = points[i];
    vector<HalfPlane<true>> planes;
    if (dir == 1) for (int i = 0, j = border.size() - 1; i <
    border.size(); j = i++) planes.emplace_back(border[j],
    border[i]);
    else for (int i = 0, j = border.size() - 1; i <
    border.size(); j = i++) planes.emplace_back(border[i],
    border[j]);
    for (int j = 0; j < n; j++) if (i != j)
    planes.emplace_back(HalfPlane<true>::getBisector(points[i],
    points[j]));
    res[i] = HPI(planes);
  }
  return res;
}

/*
hpi()는 반평면 교집합에 해당하는 다각형의 각 꼭짓점들의 좌표를
한 방향으로 정렬해서 리턴
hpi()에서 반평면 교집합의 크기가 점 하나인 경우 교집합이
없다고 판정될 수 있음.
hpi(planes, 1e-6)처럼 각 변들을 미세하게 음수방향으로 이동시킨
뒤 계산하면 됨
HalfPlane(T a, T b, T c) 생성자 사용시 isCCW는 딱히 신경 쓸
필요없음
HalfPlane(Point<T> &s, Point<T> &e) 생성자 사용할 때는 isCCW
방향 정확히 명시해줘야 함
*/

vector<Point<T>> points(n);
for (auto &[x, y] : points) cin >> x >> y;
// 볼록다각형의 꼭짓점들이 반시계방향 정렬된 경우
vector<HalfPlane<true>> planes;
for (int i = 0, j = n - 1; i < n; j = i++)
planes.emplace_back(points[j], points[i]);
// 볼록다각형의 꼭짓점들이 시계방향 정렬된 경우
vector<HalfPlane<false>> planes;
for (int i = 0, j = n - 1; i < n; j = i++)
planes.emplace_back(points[j], points[i]);
```

3.14 3D Header

```
template <typename T>
struct Point3D {
  T x, y, z;
  Point3D() = default;
  Point3D(T x, T y, T z) : x(x), y(y), z(z) {}
  template <typename U> Point3D(const Point3D<U> &other) :
  x(static_cast<T>(other.x)), y(static_cast<T>(other.y)),
  z(static_cast<T>(other.z)) {}
  bool operator<(const Point3D &other) const { return tie(x,
  y, z) < tie(other.x, other.y, other.z); }
  Point3D operator-(const Point3D &other) const { return {x -
  other.x, y - other.y, z - other.z}; }
  friend Point3D<T> cross(const Point3D<T> &p1, const
  Point3D<T> &p2) {
    return {p1.y * p2.z - p1.z * p2.y, p1.z * p2.x - p1.x *
    p2.z, p1.x * p2.y - p1.y * p2.x};
  }
  friend T dot(const Point3D<T> &p1, const Point3D<T> &p2) {
    return p1.x * p2.x + p1.y * p2.y + p1.z * p2.z;
  }
};
template <typename T>
ld getTriangleArea(const Point3D<T> &p1, const Point3D<T> &p2,
const Point3D<T> &p3) {
  auto [nx, ny, nz] = cross(p2 - p1, p3 - p1);
  return hypot<ld>(nx, ny, nz) / 2;
}
template <typename T>
tuple<T, T, T, T> getEquation(const Point3D<T> &p1, const
Point3D<T> &p2, const Point3D<T> &p3) { // ax + by + cz + d =
0
  auto [a, b, c] = cross(p2 - p1, p3 - p1);
  assert(a != 0 || b != 0 || c != 0); // a=b=c=00이면 collinear
  auto d = -(a * p1.x + b * p1.y + c * p1.z);
  return {a, b, c, d};
}
ld distPF(const Point3D<ld> &p1, const Point3D<ld> &f1, const
Point3D<ld> &f2, const Point3D<ld> &f3) { // distance from
P(point) to F(face)
  auto normal = cross(f2 - f1, f3 - f1);
  return abs<ld>(dot(normal, p1 - f1)) / hypot<ld>(normal.x,
  normal.y, normal.z);
}
```

3.15 3D Convex Hull

```
// naive O(N^4)
template <typename T>
bool isExtreme(const vector<Point3D<T>> &ps, const Point3D<T>
&p1, const Point3D<T> &p2, const Point3D<T> &p3) {
  auto normal = cross(p2 - p1, p3 - p1);
  if (normal.x == 0 && normal.y == 0 && normal.z == 0) return
  false; // collinear
  int cntPos = 0, cntNeg = 0;
  for (auto point : ps) {
    cntPos += (dot(normal, point - p1) > 0);
    cntNeg += (dot(normal, point - p1) < 0);
  }
  return cntPos == 0 || cntNeg == 0;
}
template <typename T>
vector<tuple<int, int, int>> getConvexHullFaceIdxes(const
vector<Point3D<T>> &ps) {
  vector<tuple<int, int, int>> res;
  int n = ps.size();
  for (int i = 0; i < n; ++i) {
    for (int j = i + 1; j < n; ++j) {
      for (int k = j + 1; k < n; ++k) {
        if (isExtreme(ps, ps[i], ps[j], ps[k]))
        res.emplace_back(i, j, k);
      }
    }
  }
  return res;
}

// incremental $O(N^2)$
template <typename T>
bool collinear(const Point3D<T> &p1, const Point3D<T> &p2,
const Point3D<T> &p3) {
  auto [x, y, z] = cross(p2 - p1, p3 - p1);
  return abs(x) <= 1e-9 && abs(y) <= 1e-9 && abs(z) <= 1e-9;
}
template <typename T>
vector<tuple<int, int, int>>
getConvexHullFaceIdxes(vector<Point3D<T>> ps) {
  int n = ps.size();
  assert(n >= 3);
  sort(ps.begin(), ps.end());
  vector<tuple<int, int, int>> res;
  vector<vector<bool>> edgeVisible(n, vector<bool>(n, true));
  auto add = [&ps, &res, &edgeVisible](int a, int b, int c) {
    res.emplace_back(a, b, c);
    edgeVisible[a][b] = edgeVisible[b][c] = edgeVisible[c][a]
    = false;
  };
  for (int i = 2; i < n; i++) if (!collinear(ps[0], ps[1],
  ps[i])) {
    swap(ps[2], ps[i]);
    break;
  }
  assert(!collinear(ps[0], ps[1], ps[2]) && "모든 점들이 한
  직선 위에 있음");
  add(0, 1, 2);
  add(0, 2, 1);
  for(int i = 3; i < n; i++) {
    vector<tuple<int, int, int>> invisibleFaces;
    for(auto [idx1, idx2, idx3] : res) {
      Point3D<T> normal = cross(ps[idx2] - ps[idx1], ps[idx3]
      - ps[idx1]);
      if(dot(normal, ps[i] - ps[idx1]) > 1e-9)
      edgeVisible[idx1][idx2] = edgeVisible[idx2][idx3] =
      edgeVisible[idx3][idx1] = true;
      else invisibleFaces.emplace_back(idx1, idx2, idx3);
    }
    res.clear();
    for(auto [idx1, idx2, idx3] : invisibleFaces) {
      if(edgeVisible[idx2][idx1]) add(idx2, idx1, i);
      if(edgeVisible[idx3][idx2]) add(idx3, idx2, i);
      if(edgeVisible[idx1][idx3]) add(idx1, idx3, i);
    }
    res.insert(res.end(), invisibleFaces.begin(),
    invisibleFaces.end());
```

```
  }
  return res;
}
```

3.16 Green's Theorem

반시계방향의 조각적으로 매끄러운 단순폐곡선 C와 C로 유계된 영역 D에 대해

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \int_C P\,dx + Q\,dy$$

다각형의 각 변에 대한 합으로 계산하면

$$\int_C P\,dx + Q\,dy = \sum_{i=1}^{N} \int_{(x_i,y_i)}^{(x_{i+1},y_{i+1})} P\,dx + Q\,dy$$

Examples

1. Polar Moment

$$P = -\frac{1}{3}y^3, \qquad Q = \frac{1}{3}x^3$$

$$\iint_D (x^2+y^2)\,dA = \oint_C \frac{1}{3}x^3\,dy - \frac{1}{3}y^3\,dx$$

$$= \sum_{i=1}^{N} \int_{(x_i,y_i)}^{(x_{i+1},y_{i+1})} \frac{1}{3}x^3\,dy - \frac{1}{3}y^3\,dx$$

각 변에서

$$x = x_i + (x_{i+1}-x_i)t, \qquad y = y_i + (y_{i+1}-y_i)t, \qquad 0 \le t \le 1$$

$$dx = (x_{i+1}-x_i)dt, \qquad dy = (y_{i+1}-y_i)dt$$

$$= \frac{1}{3}\sum_{i=1}^{N} \int_0^1 \left((x_i + (x_{i+1}-x_i)t)^3(y_{i+1}-y_i) - (y_i + (y_{i+1}-y_i)t)^3(x_{i+1}-x_i)\right) dt$$

$$= \frac{1}{12}\sum_{i=1}^{N} (x_i y_{i+1} - x_{i+1}y_i)\left(x_i^2 + x_i x_{i+1} + x_{i+1}^2 + y_i^2 + y_i y_{i+1} + y_{i+1}^2\right)$$

2. First Moment

$$\iint_D x\,dA = \oint_C \frac{1}{2}x^2\,dy = \frac{1}{6}\sum_{i=1}^{N} (y_{i+1}-y_i)(x_i^2 + x_i x_{i+1} + x_{i+1}^2)$$

Area

$$S = \iint_D 1\,dA$$

$$P = -\frac{y}{2}, \qquad Q = \frac{x}{2}, \qquad \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) = 1$$

$$\therefore S = \int_C (P\,dx + Q\,dy) = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

Circular Arc Integral

$$x = x_c + r\cos\theta, \qquad y = y_c + r\sin\theta$$

$$S = \frac{1}{2}\oint (x\,dy - y\,dx) = \frac{r}{2}\left[x_c\sin\theta - y_c\cos\theta + r\theta\right]_{\theta_1}^{\theta_2}$$

Line Segment Integral

$$v_x = x_2 - x_1, \qquad v_y = y_2 - y_1$$

$$x = x_1 + v_x t, \qquad y = y_1 + v_y t, \qquad 0 \le t \le 1$$

$$S = \frac{1}{2}\oint (x\,dy - y\,dx) = \frac{1}{2}\left[(v_y x_1 - v_x y_1)t\right]_0^1$$

$$= \frac{1}{2}(y_2x_1 - y_1x_1 - x_2y_1 + x_1y_1) = \frac{1}{2}(x_1y_2 - x_2y_1)$$

3.17 Geometry Theory

(x,y)의 (a,b) 기준 대칭이동 : $((a^2-b^2)(-x) - 2aby, (a^2-b^2)(y) - 2abx)$

Barycentric Coordinates

$\overline{A(x_a,y_a)}$, $\overline{B(x_b,y_b)}$, $\overline{C(x_c,y_c)}$인 $\triangle ABC$에서
$\overline{BC} = a, \overline{CA} = b, \overline{AB} = c, \mathcal{A} = b^2 + c^2 - a^2, \mathcal{B} = c^2 + a^2 - b^2, \mathcal{C} = a^2 + b^2 - c^2$라 할 때 오심의 Barycentric 좌표 $(\alpha : \beta : \gamma)$
외심 $(\alpha : \beta : \gamma) = (a^2\mathcal{A} : b^2\mathcal{B} : c^2\mathcal{C})$
내심 $(\alpha : \beta : \gamma) = (a : b : c)$
무게중심 $(\alpha : \beta : \gamma) = (1 : 1 : 1)$
수심 $(\alpha : \beta : \gamma) = (\mathcal{BC} : \mathcal{CA} : \mathcal{AB})$
(꼭짓점 A 맞은편의) 방심 $(\alpha : \beta : \gamma) = (-a : b : c)$
직교좌표계로 표현하면 $\left(\frac{\alpha x_a + \beta x_b + \gamma x_c}{\alpha+\beta+\gamma}, \frac{\alpha y_a + \beta y_b + \gamma y_c}{\alpha+\beta+\gamma}\right)$

3D Rotation

$$\begin{pmatrix} v_x^2(1-c)+c & v_xv_y(1-c)-v_zs & v_xv_z(1-c)+v_ys \\ v_xv_y(1-c)+v_zs & v_y^2(1-c)+c & v_yv_z(1-c)-v_xs \\ v_xv_z(1-c)-v_ys & v_yv_z(1-c)+v_xs & v_z^2(1-c)+c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

4 Graph

4.1 Bellman-Ford

```
auto bellmanFord = [&](int s) { // O(VE)
  vector<dist_t> dist(n + 1, INF);
  dist[s] = 0;
  for (int _ = n - 1; _--;) {
    for (auto [u, v, w] : edges) if (dist[u] != INF) dist[v] =
    min(dist[v], dist[u] + w);
  }
  bool hasNegCycle = false;
  for (auto [u, v, w] : edges) hasNegCycle |= (dist[u] != INF
  && dist[v] > dist[u] + w);
  return pair{hasNegCycle, dist};
};
```

4.2 SPFA

```
auto spfa = [&](int s) { // O(VE) // average O(V+E)
  vector<dist_t> dist(n + 1, INF);
  dist[s] = 0;
  queue<int> q;
  q.push(s);
  vector<int> cnt(n + 1);
  ++cnt[s];
  vector<bool> inq(n + 1);
  while (!q.empty()) {
    auto cur = q.front();
    q.pop();
    inq[cur] = false;
    for (auto [nxt, cost] : adj[cur]) if (dist[nxt] >
    dist[cur] + cost) {
      dist[nxt] = dist[cur] + cost;
      if (inq[nxt]) continue;
      if (++cnt[nxt] >= n) return pair{true, dist}; //
      hasNegCycle
      q.push(nxt);
      inq[nxt] = true;
    }
  }
  return pair{false, dist}; // {hasNegCycle, dist}
};
```

4.3 SCC

```
auto getSCC(int n, const vector<vector<int>> &adj) { // adj:
0-based // O(V+E)
  vector<int> dfsn(n), sccn(n, -1);
  vector<int> s(n);
  int top = 0, dfsi = 0, scci = 0;
  auto dfs = [&](auto &&dfs, int cur) -> int {
    int low = dfsn[cur] = ++dfsi;
    s[top++] = cur;
    for (auto nxt : adj[cur]) if (!~sccn[nxt]) low = min(low,
    dfsn[nxt] ? dfsn[nxt] : dfs(dfs, nxt));
    if (low == dfsn[cur]) {
      do { sccn[s[--top]] = scci; } while (s[top] != cur);
      ++scci;
    }
    return low;
  };
  for (int i = 0; i < n; i++) if (!dfsn[i]) dfs(dfs, i);
  for (int i = 0; i < n; i++) sccn[i] = scci - 1 - sccn[i];
  return pair{scci, sccn}; // 0-based // scci = scc 개수
}
// 1-based인 경우 사용법
```

```
// auto [scci, sccn] = getSCC(n + 1, adj);
// --scci; // 정점 0 제외

// graphToDAG(n, adj, sccn) // 0-based
//     vector<vector<int>> sccs(scci);
//     for (int i = 0; i < n; i++) sccs[sccn[i]].push_back(i);
//     vector<int> visited(sccs.size(), -1);
//     vector<vector<int>> dag(sccs.size());
//     for (auto &scc : sccs) for (auto u : scc) {
//       for (auto v : adj[u]) if (sccn[v] != sccn[u] &&
visited[sccn[v]] != sccn[u]) {
//         visited[sccn[v]] = sccn[u];
//         dag[sccn[u]].push_back(sccn[v]);
//       }
//     }
```

4.4 2CC

```
vector<int> dfsn(n), eccn(n), s(n);
int dfsi = 0, ecci = 0, top = 0;
auto dfs = [&](auto &&dfs, int cur, int pid) -> int {
    int low = dfsn[cur] = ++dfsi;
    s[top++] = cur;
    for (auto [nxt, id] : adj[cur]) if (id != pid) low =
    min(low, dfsn[nxt] ?: dfs(dfs, nxt, id));
    if (low == dfsn[cur]) {
        do { eccn[s[--top]] = ecci; } while (s[top] != cur);
        ++ecci;
    }
    return low;
};
for (int i = 0; i < n; i++) if (!dfsn[i]) dfs(dfs, i, -1);
```

4.5 BCC

```
auto get_BCC(int n, int m, const vector<vector<pair<int,
int>>> &adj) { // O(V + E)
    vector<vector<int>> bccs;
    vector<int> dfsn(n + 1, 0), s(m);
    int dfsi = 0, top = 0;
    auto dfs = [&](auto &&dfs, int cur, int pid) -> int {
        int low = dfsn[cur] = ++dfsi;
        for (auto [nxt, id] : adj[cur]) if (id != pid) {
            if (!dfsn[nxt]) {
                s[top++] = id;
                int nlow = dfs(dfs, nxt, id);
                low = min(low, nlow);
                if (nlow >= dfsn[cur]) {
                    vector<int> bcc;
                    do { bcc.push_back(s[--top]); } while
                    (s[top] != id);
                    bccs.push_back(bcc);
                }
            }
            else {
                if (dfsn[nxt] < dfsn[cur]) s[top++] = id;
                low = min(low, dfsn[nxt]);
            }
        }
        return low;
    };
    for (int i = 1; i <= n; i++) if (!dfsn[i]) dfs(dfs, i,
    -1);
    return bccs;
}
vector<int> getNodeOfBCC(int n, const vector<int> &bcc, const
vector<pair<int, int>> &edges) {
    static int t = 1;
    static vector<int> visited(n + 1);
    if (visited.size() < n + 1) visited.resize(n + 1); //
    테스트케이스 여러 개인 문제면 n 바뀔 수 있음
    vector<int> res;
    for (auto id : bcc) {
        auto [u, v] = edges[id];
        if (visited[u] != t) res.push_back(u);
        if (visited[v] != t) res.push_back(v);
        visited[u] = visited[v] = t;
    }
    ++t;
    return res;
    // int E = bcc.size(), V = getNodeOfBCC(n, bcc).size();
    // E = V : 모든 정점의 차수가 2인 사이클(simple graph면
    단순사이클)
    // E > V : 독립적인 사이클 2개 이상 존재
    // E < V : 단절선(E=1, V=2) 뿐임
    // E=0, V=1 : 정점하나만 동떨어진 경우
}
// 2개 이상의 BCC에 포함되는 정점은 단절점이다.(역도 성립)
// $V=2$, $E=1$라면 해당 BCC는 단절선이다.(역도 성립)
```

4.6 Articulation Points

```
vector<int> dfsn(n + 1);
int dfsi = 0;
auto dfs = [&](auto &&dfs, int cur, int parent) -> int {
  int low = dfsn[cur] = ++dfsi, child = 0;
  bool cutP = 0;
  for (auto nxt : adj[cur]) if (nxt != parent) {
    if (!dfsn[nxt]) {
      ++child;
      int nlow = dfs(dfs, nxt, cur);
      low = min(low, nlow);
      cutP |= nlow >= dfsn[cur];
    }
    else low = min(low, dfsn[nxt]);
  }
  if (!~parent ? child > 1 : cutP) ; // cur은 단절점
  return low;
};
for (int i = 1; i <= n; i++) if(!dfsn[i]) dfs(dfs, i, -1);
```

4.7 Bridges

```
vector<int> dfsn(n + 1);
int dfsi = 0;
auto dfs = [&](auto &&dfs, int cur, int pid) -> int {
  int low = dfsn[cur] = ++dfsi;
  for (auto [nxt, id] : adj[cur]) if (id != pid) {
    if (!dfsn[nxt]) {
      int nlow = dfs(dfs, nxt, id);
      low = min(low, nlow);
      if (nlow > dfsn[cur]) ; // id: bridge
    }
    else low = min(low, dfsn[nxt]);
  }
  return low;
};
for (int i = 1; i <= n; i++) if(!dfsn[i]) dfs(dfs, i, -1);
```

4.8 2-SAT

```
struct TwoSat {
  int n;
  vector<vector<int>> adj;
  vector<int> sccn;
  TwoSat(int n) : n(n), adj(2 * n) {}
  int f(int a) { // 1-based->0-based
    if (a > 0) return 2 * (a - 1);
    return 2 * (-a - 1) + 1;
  }
  // 전부 1-based
  void add2SAT(int a, int b) { // a or b가 true
    adj[f(-a)].push_back(f(b));
    adj[f(-b)].push_back(f(a));
  }
  void add1SAT(int a) { adj[f(-a)].push_back(f(a)); } // a가
  true
  void addEqual2SAT(int a, int b) { add2SAT(a, -b);
  add2SAT(-a, b); } // a == b 가 true
  void addNotEqual2SAT(int a, int b) { add2SAT(a, b);
  add2SAT(-a, -b); } // a != b 가 true // a, b 중 정확히
  하나만 true
  void add22SAT(int a1, int a2, int b1, int b2) { add2SAT(a1,
  b1); add2SAT(a1, b2); add2SAT(a2, b1); add2SAT(a2, b2); } //
  (a1 and a2) or (b1 and b2)가 true
  void addNMSAT(const vector<int> &as, const vector<int> &bs)
  { for (auto &a : as) for (auto &b : bs) add2SAT(a, b); } //
  (a1 and a2 and ... and aN) or (b1 and b2 and ... and bM)가
  true
  void addAtLeastTwo(int a, int b, int c) { add2SAT(a, b);
  add2SAT(b, c); add2SAT(c, a); } // a, b, c 중 2개 이상이
  true
  void addAtMostOne(const vector<int> &v) { // v원소 중 최대
  하나만 true
    if (v.size() <= 1) return;
    adj.resize(2 * (n + v.size()));
    for (int i = 0; i < v.size(); i++) add2SAT(-v[i], n + 1 +
    i);
    for (int i = 1; i < v.size(); i++) {
      int cur = n + 1 + i;
      int prv = n + i;
      add2SAT(-prv, cur);
```

```
      add2SAT(-prv, -v[i]);
    }
    n += v.size();
  }
  bool solve() { // O(V + E) // V는 변수개수, E는 절 개수
    sccn = getSCC(2 * n, adj).second;
    for (int i = 1; i <= n; i++) if (sccn[f(i)] ==
    sccn[f(-i)]) return false;
    return true;
  }
  bool getSolution(int i) { // 1-based
    return sccn[f(-i)] < sccn[f(i)];
  }
};
// false/true는 -/+로 표현
// graph.add2SAT(-1, 2); // (!x1 or x2)
// graph.add2SAT(-2, -3); // (!x2 or !x3)
```

4.9 Euler Circuit

```
// 양방향
struct Edge {
  int to, rev, cnt;
};

auto dfs = [&](auto &&dfs, int cur) -> void {
  while (!adj[cur].empty()) {
    auto &edge = adj[cur].back();
    if (edge.cnt == 0) adj[cur].pop_back();
    else {
      --edge.cnt;
      --adj[edge.to][edge.rev].cnt;
      int tmp = edge.idx;
      dfs(dfs, edge.to);
      edges.push_back(tmp);
    }
  }
  nodes.push_back(cur);
};

// 양방향 그래프
struct Edge { int to, rev, cnt; };
pair<bool, vector<int>> getCircuit(vector<vector<Edge>> adj,
int start) { // adj비워져도 괜찮으면 참조 사용 // O(V+E)
  int E = 0;
  for (auto &r : adj) for (auto edge : r) E += edge.cnt;
  E >>= 1;
  vector<int> res;
  auto dfs = [&](auto &&dfs, int cur) -> void {
    while (!adj[cur].empty()) {
      auto &edge = adj[cur].back();
      if (edge.cnt == 0) adj[cur].pop_back();
      else {
        --edge.cnt;
        --adj[edge.to][edge.rev].cnt;
        dfs(dfs, edge.to);
      }
    }
    res.push_back(cur);
  };
  dfs(dfs, start);
  bool possible = (res.size() == E + 1 && res[0] ==
  res.back());
  return {possible, res};
}

// 단방향 그래프
pair<bool, vector<int>> getCircuit(vector<vector<pair<int,
int>>> adj, int start) { // adj비워져도 괜찮으면 참조 사용 //
O(V+E)
  int E = 0;
  for (auto &r : adj) for (auto [nxt, cnt] : r) E += cnt;
  vector<int> res;
  auto dfs = [&](auto &&dfs, int cur) -> void {
    while (!adj[cur].empty()) {
      auto &[nxt, cnt] = adj[cur].back();
      int tmp = nxt; // dangling 방지
      if (--cnt == 0) adj[cur].pop_back();
      dfs(dfs, tmp);
    }
    res.push_back(cur);
  };
  dfs(dfs, start);
  reverse(res.begin(), res.end());
  bool possible = (res.size() == E + 1 && res[0] ==
  res.back());
  return {possible, res};
}

// 오일러 회로는 모든 간선을 지나는 게 목표지, 모든 정점을
지나는 게 목표가 아님
// 따라서 간선들끼리만 연결되어있으면 가능하고, start는 간선과
연결되어 있는 정점 아무거나 가능

// 오일러 경로는 홀수 차수 정점이 2개 or 0개여야 가능.
// 만약 2개라면 해당정점들을 cnt=1인 간선으로 잇고 회로를 찾은
뒤 해당간선만 제거하면 됨

// 경로 구하고 싶으면 indg < outdeg인 지점을 start로 호출하면
됨, 이 경우 res[0] != res.back()이 됨
```

4.10 C3, C4

```
void getC3(const vector<vector<int>> &adj) { // O(m sqrt(m))
  vector<vector<int>> dag(adj.size());
  for (int u = 0; u < adj.size(); u++) {
    for (auto v : adj[u]) {
      if (adj[u].size() < adj[v].size() || (adj[u].size() ==
      adj[v].size() && u < v)) dag[u].push_back(v);
    }
  } // dag에서 outdegree는 전부 O(sqrt(m)) 이하
  vector<int> mark(adj.size(), -1);
  for (int i = 0; i < adj.size(); i++) {
    for (auto j : dag[i]) mark[j] = i;
    for (auto j : dag[i]) {
      for (auto k : dag[j]) {
        if (mark[k] == i) cout << i << " " << j << " " << k <<
        "\n";
      }
    }
  }
}
// rank(i)<rank(j)<rank(k)
int getC4(const vector<vector<int>> &adj, int MOD) {
  auto cmp = [&](int u, int v) { // u < v
    return adj[u].size() < adj[v].size() || (adj[u].size() ==
    adj[v].size() && u < v);
  };
  vector<vector<int>> dag(adj.size());
  for (int u = 0; u < adj.size(); u++) {
    for (auto v : adj[u]) if (cmp(u, v)) dag[u].push_back(v);
  }
  int res = 0;
  vector<int> cnt(adj.size());
  for (int i = 0; i < adj.size(); i++) {
    for (auto j : adj[i]) {
      for (auto k : dag[j]) if (cmp(i, k)) {
        res += cnt[k];
        if (res >= MOD) res -= MOD;
        ++cnt[k];
      }
    }
    for (auto j : adj[i]) {
      for (auto k : dag[j]) if (cmp(i, k)) cnt[k] = 0;
    }
  }
  return res;
}
// rank(i), rank(j), rank(j') < rank(k)
```

4.11 PEO

```
pair<bool, vector<int>> getPEO(int n, const
vector<vector<int>>& adj) { // O(V+E) // 1-based
  vector<int> label(n + 1), order;
  vector<bool> visited(n + 1, false);
  vector<vector<int>> buckets(n + 1);
  for (int i = 1; i <= n; i++) buckets[0].push_back(i);
  int mx = 0;
  while (order.size() < n) { // mcs
    while (buckets[mx].empty()) --mx;
    int cur = buckets[mx].back();
    buckets[mx].pop_back();
    if (visited[cur]) continue;
    visited[cur] = true;
    order.push_back(cur);
    for (auto nxt : adj[cur]) if (!visited[nxt]) {
      buckets[++label[nxt]].push_back(nxt);
      mx = max(mx, label[nxt]);
```

```
    }
  }
  reverse(order.begin(), order.end());
  // check peo
  vector<int> pos(n + 1), parent(n + 1, -1);
  for (int i = 0; i < n; i++) pos[order[i]] = i;
  vector<vector<int>> chd(n + 1);
  for (int u = 1; u <= n; u++) {
    int mn = n + 1, w = -1;
    for (auto v : adj[u]) if (pos[v] > pos[u] && mn > pos[v])
    mn = pos[v], w = v;
    if (~w) {
      parent[u] = w;
      chd[w].push_back(u);
    }
  }
  vector<int> tag(n + 1);
  for (auto u : order) {
    tag[u] = u;
    for (auto v : adj[u]) tag[v] = u;
    for (auto v : chd[u]) {
      for (auto w : adj[v]) {
        if (pos[w] > pos[v] && w != u) {
          if (tag[w] != u) return {false, order};
        }
      }
    }
  }
  return {true, order};
} // {isChordal, peo} // chordal: 길이 4이상의 모든 simple
cycle이 chord를 포함
```

4.12 Bimatch

```
// addEdge(l, r) : adj[l].push_back(r); // 0<=l<n1, 0<=r<n2
auto bimatch(int n1, int n2, const vector<vector<int>> &adj) {
// O(VE) // 0-based
  vector<int> ml(n1, -1), mr(n2, -1), vr(n2);
  int t = 1;
  auto dfs = [&](auto &&dfs, int l) -> bool {
    for (auto r : adj[l]) if (vr[r] != t) {
      vr[r] = t;
      if (!~mr[r] || dfs(dfs, mr[r])) {
        ml[l] = r;
        mr[r] = l;
        return true;
      }
    }
    return false;
  };
  int res = 0;
  for (int l = 0; l < n1; ++l, ++t) res += dfs(dfs, l);
  return tuple{res, ml, mr};
}
auto get_mvc(int n1, int n2, const vector<vector<int>> &adj,
const vector<int> &ml, const vector<int> &mr) { // O(V+E) //
0-based
  vector<bool> vl(n1), vr(n2);
  auto dfs = [&](auto &&dfs, int l) -> void {
    vl[l] = true;
    for (auto r : adj[l]) if (~mr[r] && !vr[r] && !vl[mr[r]])
    {
      vr[r] = true;
      dfs(dfs, mr[r]);
    }
  };
  for (int l = 0; l < n1; l++) if (!~ml[l]) dfs(dfs, l);
  vector<int> mvcL, mvcR;
  for (int l = 0; l < n1; l++) if (!vl[l]) mvcL.push_back(l);
  for (int r = 0; r < n2; r++) if (vr[r]) mvcR.push_back(r);
  return pair{mvcL, mvcR};
}
```

4.13 Hopcroft Karp

```
auto bimatch(int n1, int n2, const vector<vector<int>> &adj) {
// O(E sqrt(V)) // 0-based
  vector<int> ml(n1, -1), mr(n2, -1), lvl(n1), ptr(n1);
  auto bfs = [&]() -> bool {
    queue<int> q;
    for (int l = 0; l < n1; l++) {
      if (!~ml[l]) {
        lvl[l] = 0;
        q.push(l);
      }
      else lvl[l] = -1;
    }
    bool flag = false;
    while (!q.empty()) {
      int l = q.front();
      q.pop();
      for (auto r : adj[l]) {
        if (!~mr[r]) flag = true;
        else if (!~lvl[mr[r]]) {
          lvl[mr[r]] = lvl[l] + 1;
          q.push(mr[r]);
        }
      }
    }
    return flag;
  };
  auto dfs = [&](auto &&dfs, int l) -> bool {
    for (int &i = ptr[l]; i < adj[l].size(); i++) {
      int r = adj[l][i];
      if (!~mr[r] || lvl[mr[r]] == lvl[l] + 1 && dfs(dfs,
      mr[r])) {
        ml[l] = r;
        mr[r] = l;
        return true;
      }
    }
    return false;
  };
  int res = 0;
  while (bfs()) {
    fill(ptr.begin(), ptr.end(), 0);
    for (int l = 0; l < n1; l++) res += (!~ml[l] && dfs(dfs,
    l));
  }
  return tuple{res, ml, mr};
}
```

4.14 Dinic

```
template <typename F>
struct Dinic {
  struct Edge {
    int to, rev;
    F c, oc;
    F flow() { return max(oc - c, F(0)); }
  };
  vector<int> lvl, ptr, q;
  vector<vector<Edge>> adj;
  Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {} // 0-based
  void addEdge(int a, int b, F c, F rcap = 0) { // 양방향이면
  rcap=c로 호출
    if (a == b) return; // self-loop는 최대유량에 영향 X
    adj[a].push_back({b, adj[b].size(), c, c});
    adj[b].push_back({a, adj[a].size() - 1, rcap, rcap});
  }
  F dfs(int cur, int t, F mn) {
    if (cur == t || !mn) return mn;
    for (int &i = ptr[cur]; i < adj[cur].size(); i++) {
      auto &e = adj[cur][i];
      if (lvl[e.to] != lvl[cur] + 1) continue;
      if (F f = dfs(e.to, t, min(mn, e.c))) {
        e.c -= f, adj[e.to][e.rev].c += f;
        return f;
      }
    }
    return 0;
  }
  template <bool scaling=false> F maxFlow(int s, int t) { //
  O(V^2 E) // max(cap)이 큰 경우 scaling=true가 빠름
    F res = 0; q[0] = s;
    for (int i = scaling ? 0 : 30; i < 31; i++) do {
      lvl = ptr = vector<int>(q.size());
      int qs = 0, qe = lvl[s] = 1;
      while (qs < qe && !lvl[t]) {
        int cur = q[qs++];
        for (Edge &e : adj[cur]) if (!lvl[e.to] && e.c >> (30
        - i)) {
          q[qe++] = e.to, lvl[e.to] = lvl[cur] + 1;
        }
      }
      while (F f = dfs(s, t, numeric_limits<F>::max())) res +=
      f;
    } while (lvl[t]);
    return res;
```

```
  }
  bool leftOfMinCut(int a) { return lvl[a] != 0; }
};
// vector<pair<int, int> ref;
// ref.emplace_back(u, graph.adj[u].size()); graph.addEdge(u,
v);
// auto [u, idx] = ref[i]; cout << graph.adj[u][idx].flow() <<
"\n";
// O(min(Ef, V^2 E)), all cap = 1: O(min(V^{2/3}, E^{1/2})E)
// dinic with scaling O(VE log(max_cap))
// 근데 보통 그냥 dinic이 더 빠름, 오히려 max_cap이 클 때
dinic with scaling이 좋음
// 무한간선 많은 경우 proxysource -> source로 K 용량의 간선을
두고 계산하면 애초에 최대유량이 K만큼으로 제한되므로
오버플로우 방지 가능
// 정점분할 시 양방향이라면 addEdge(out(a), in(b), c),
addEdge(out(b), int(a), c)
// 정점분할 시 maxFlow(out(s), in(t))
```

4.15 MCMF

```
template <typename C, typename F>
struct Graph {
  struct Edge {
    int to, rev;
    F c, oc;
    C cost;
    int from;
  };
  vector<vector<Edge>> adj;
  C DIST_INF = numeric_limits<C>::max();
  Graph(int n) : adj(n) {} // 0-based
  void addEdge(int a, int b, C cost, F c) {
    adj[a].push_back({b, adj[b].size(), c, c, cost, a});
    adj[b].push_back({a, adj[a].size() - 1, 0, 0, -cost, b});
  }
  pair<C, F> mcmf(int s, int t, F targetFlow =
  numeric_limits<F>::max()) { // O(VEf) // average O(Ef)
    C minCost = 0;
    F maxFlow = 0;
    vector<Edge *> pedge(adj.size());
    vector<C> dist(adj.size());
    vector<bool> inq(adj.size());
    while (maxFlow < targetFlow) {
      fill(dist.begin(), dist.end(), DIST_INF);
      dist[s] = 0;
      queue<int> q;
      q.push(s);
      inq[s] = true;
      while (!q.empty()) {
        int cur = q.front();
        q.pop();
        inq[cur] = false;
        for (auto &e : adj[cur]) if (e.c && dist[e.to] >
        dist[cur] + e.cost) {
          dist[e.to] = dist[cur] + e.cost;
          pedge[e.to] = &e;
          if (!inq[e.to]) {
            q.push(e.to);
            inq[e.to] = true;
          }
        }
      }
      if (dist[t] == DIST_INF) break;
      F f = targetFlow - maxFlow;
      for (Edge *e = pedge[t]; e; e = pedge[e->from]) f =
      min(f, e->c);
      for (Edge *e = pedge[t]; e; e = pedge[e->from]) {
        e->c -= f;
        adj[e->to][e->rev].c += f;
      }
      minCost += f * dist[t];
      maxFlow += f;
    }
    return {minCost, maxFlow};
  }
};
```

4.16 MCMF(dijkstra)

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/priority_queue.hpp>
template <typename C, typename F>
struct Graph {
  struct edge {
    int to, rev;
    F c, oc;
    C cost;
    int from;
  };
  int n;
  vector<vector<edge>> adj;
  vector<int> seen;
  vector<C> dist, pi;
  vector<edge*> pedge;
  const C DIST_INF = numeric_limits<C>::max() / 4;
  Graph(int n) : n(n), adj(n), seen(n), dist(n), pi(n),
  pedge(n) {} // 0-based
  void addEdge(int a, int b, C cost, F c) {
    adj[a].push_back({b, adj[b].size(), c, c, cost, a});
    adj[b].push_back({a, adj[a].size() - 1, 0, 0, -cost, b});
  }
  void path(int s) {
    fill(seen.begin(), seen.end(), 0);
    fill(dist.begin(), dist.end(), DIST_INF);
    dist[s] = 0;
    using elem = pair<C, int>;
    using pq_t = __gnu_pbds::priority_queue<elem,
    greater<elem>>;
    pq_t q;
    vector<typename pq_t::point_iterator> its(n);
    q.push({0, s});
    while (!q.empty()) {
      int cur = q.top().second;
      q.pop();
      seen[cur] = 1;
      C di = dist[cur] + pi[cur];
      for (auto &e : adj[cur]) if (e.c && !seen[e.to]) {
        C val = di - pi[e.to] + e.cost;
        if (dist[e.to] > val) {
          dist[e.to] = val;
          pedge[e.to] = &e;
          if (its[e.to] == q.end()) its[e.to] =
          q.push({dist[e.to], e.to});
          else q.modify(its[e.to], {dist[e.to], e.to});
        }
      }
    }
    for (int i = 0; i < n; i++) pi[i] = min(pi[i] + dist[i],
    DIST_INF);
  }
  pair<C, F> mcmf(int s, int t) {
    C minCost = 0;
    F maxFlow = 0;
    while (path(s), seen[t]) {
      F f = numeric_limits<F>::max();
      for (edge *e = pedge[t]; e; e = pedge[e->from]) f =
      min(f, e->c);
      for (edge *e = pedge[t]; e; e = pedge[e->from]) {
        e->c -= f;
        adj[e->to][e->rev].c += f;
      }
      maxFlow += f;
      minCost += f * (pi[t] - pi[s]);
    }
    return {minCost, maxFlow};
  }
  void setpi(int s) { // 음수비용 존재 시 mcmf전에 호출필요 //
  O(VE)
    fill(pi.begin(), pi.end(), DIST_INF);
    pi[s] = 0;
    int it = n, ch = 1; ll v;
    while (ch-- && it--)
      for (int i = 0; i < n; i++) if (pi[i] != DIST_INF)
        for (edge& e : adj[i]) if (e.c)
          if ((v = pi[i] + e.cost) < pi[e.to])
            pi[e.to] = v, ch = 1;
    assert(it >= 0); // 비용 음수사이클 존재
  }
};
```

4.17 LCA

```
// 1-based (ac default값이 0이라서 0-based안됨)
int mx = __lg(n);
vector<int> dep(n + 1);
vector ac(n + 1, vector<int>(mx + 1));
auto dfs = [&](auto &&dfs, int cur, int par) -> void {
  for (auto nxt : adj[cur]) if (nxt != par) {
```

```
    dep[nxt] = dep[cur] + 1;
    ac[nxt][0] = cur;
    for (int i = 1; i <= mx; i++) ac[nxt][i] = ac[ac[nxt][i -
    1]][i - 1];
    dfs(dfs, nxt, cur);
  }
};
dfs(dfs, 1, -1); // O(NlogN)
auto getLCA = [&](int a, int b) { // O(logN)
  if (dep[a] > dep[b]) swap(a, b);
  for (int diff = dep[b] - dep[a]; diff; diff &= diff - 1) b =
  ac[b][__builtin_ctz(diff)];
  if (a == b) return a;
  for (int i = mx; i >= 0; i--) if (ac[a][i] != ac[b][i]) a =
  ac[a][i], b = ac[b][i];
  return ac[a][0];
};
auto getLCA2 = [&](int a, int b, int root) {
  int x = getLCA(root, a);
  int y = getLCA(root, b);
  int res = getLCA(a, b);
  if (dep[res] < dep[x]) res = x;
  if (dep[res] < dep[y]) res = y;
  return res;
};
```

4.18 LCA(RMQ)

```
vector<int> dep(n + 1), pos(n + 1), euler(2 * n - 1);
int ei = 0;
auto mkt = [&](auto &&mkt, int cur, int par) -> void {
  euler[pos[cur] = ei++] = cur;
  for (auto nxt : adj[cur]) if (nxt != par) {
    dep[nxt] = dep[cur] + 1;
    mkt(mkt, nxt, cur);
    euler[ei++] = cur;
  }
};
mkt(mkt, 1, -1);
vector<vector<int>> ac(__lg(ei) + 1, euler);
for (int i = 1; i <= __lg(ei); i++) {
  for (int j = 0; j + (1 << i) - 1 < ei; j++) {
    int u = ac[i - 1][j];
    int v = ac[i - 1][j + (1 << i - 1)];
    ac[i][j] = dep[u] < dep[v] ? u : v;
  }
}
auto getLCA = [&](int a, int b) {
  auto [l, r] = minmax(pos[a], pos[b]);
  int i = __lg(r - l + 1);
  int u = ac[i][l];
  int v = ac[i][r - (1 << i) + 1];
  return dep[u] < dep[v] ? u : v;
};
```

4.19 ETT

```
// root->v path add / node v qry
// upd(in[v], out[v], add) / qry(in[v], out[v])
// node v add / root->v path qry
// upd(in[v], out[v], add]) / qry(in[v])
// 루트가 r로 바뀌었을 때 v의 subtree
// v == r : 전체 트리
// v가 r의 조상이 아님 : 기존 subtree(v) = [tin[v], tout[v]]
// v가 r의 조상임 : 전체에서, v 아래에서 r로 가는 child의
subtree를 뺀 것
```

4.20 HLD

```
vector<vector<int>> chd(n);
vector<int> sz(n, 1), dep(n), p(n), in(n), top(n);
auto mkt = [&](auto &&mkt, int cur, int par) -> void {
  for (auto nxt : adj[cur]) if (nxt != par) {
    p[nxt] = cur;
    dep[nxt] = dep[cur] + 1;
    mkt(mkt, nxt, cur);
    chd[cur].push_back(nxt);
    sz[cur] += sz[nxt];
    if (sz[chd[cur][0]] < sz[nxt]) swap(chd[cur][0],
    chd[cur].back());
  }
};
mkt(mkt, root, -1);
int dfsi = 0;
auto ett = [&](auto &&ett, int cur) -> void {
  in[cur] = ++dfsi;
  for (auto nxt : chd[cur]) {
    top[nxt] = (nxt == chd[cur][0] ? top[cur] : nxt);
    ett(ett, nxt);
  }
};
top[root] = root;
ett(ett, root);
segtree seg(n);
auto qry = [&](int a, int b) { // O(log^2N)
  int res = 0;
  while (top[a] != top[b]) {
    if (dep[top[a]] > dep[top[b]]) swap(a, b);
    res += seg.qry(in[top[b]], in[b]);
    b = p[top[b]];
  }
  if (dep[a] > dep[b]) swap(a, b);
  res += seg.qry(in[a], in[b]); // 정점쿼리
  // if (in[a] + 1 <= in[b]) res += seg.qry(in[a] + 1, in[b]);
  // 간선쿼리
  return res;
};
// seg.upd(in[a], b);
// ett: [in[cur], in[cur] + sz[cur] - 1]
```

4.21 Centroid Decomposition

```
vector<int> sz(n + 1);
vector<bool> rm(n + 1);
auto get_sz = [&](auto &&get_sz, int cur, int par) -> int {
  sz[cur] = 1;
  for (auto nxt : adj[cur]) if (nxt != par && !rm[nxt]) {
    sz[cur] += get_sz(get_sz, nxt, cur);
  }
  return sz[cur];
};
auto get_ct = [&](auto &&get_ct, int cur, int par, int tot) ->
int {
  for (auto nxt : adj[cur]) if (nxt != par && !rm[nxt]) {
    if (sz[nxt] * 2 > tot) return get_ct(get_ct, nxt, cur,
    tot);
  }
  return cur;
};

ll ans = 0;
vector<int> a(n + 1);
vector<int> dirty;
auto f = [&](auto &&f, int cur, int par) -> void {
  // ans <- cur 기여분만큼 업데이트하기
  for (auto nxt : adj[cur]) if (nxt != par && !rm[nxt]) f(f,
  nxt, cur);
};
auto upd = [&](auto &&upd, int cur, int par) -> void {
  // a[cur];
  // dirty.push_back(cur);
  for (auto nxt : adj[cur]) if (nxt != par && !rm[nxt])
  upd(upd, nxt, cur);
};
auto dnc = [&](auto &&dnc, int cur) -> void {
  int tot = get_sz(get_sz, cur, -1);
  int ct = get_ct(get_ct, cur, -1, tot);
  rm[ct] = 1;
  for (auto nxt : adj[ct]) if (!rm[nxt]) {
    f(f, nxt, -1);
    upd(upd, nxt, -1);
  }
  // centroid가 끝점인 경로도 세야됨
  for (auto x : dirty) a[x] = 0;
  dirty.clear();
  for (auto nxt : adj[ct]) if (!rm[nxt]) dnc(dnc, nxt);
};
```

4.22 Virtual Tree

```
// compress
sort(nodes.begin(), nodes.end(), [&](int a, int b) {
  return dfsn[a] < dfsn[b];
});
int tmp = nodes.size();
for (int i = 1; i < tmp; i++) nodes.push_back(getLCA(nodes[i -
1], nodes[i]));
sort(nodes.begin(), nodes.end(), [&](int a, int b) {
  return dfsn[a] < dfsn[b];
});
nodes.erase(unique(nodes.begin(), nodes.end()), nodes.end());
root = nodes[0];
```

```
for (int i = 1; i < nodes.size(); i++) {
  int u = getLCA(nodes[i - 1], nodes[i]);
  int v = nodes[i];
  chd[u].emplace_back(v, dist[v] - dist[u]);
}

// clean-up // O(sum(nodes.size()))
for (auto x : nodes) {
  chd[x].clear();
}
```

4.23 Graph Theory

트리의 중심

모든 트리는 1개 or 2개의 중심을 가지며 2개라면 그 두 정점은 반드시 인접함
트리의 모든 지름은 반드시 트리의 중심을 지남(=공유함)
트리의 지름이 짝수라면: 중심은 1개, 지름의 중간지점이 중심
홀수라면: 중심은 2개, 지름의 가운데 간선으로 연결된 두 정점이 중심
즉, 중심이 1개라면 모든 지름은 그 중심을 지나고 2개라면 모든 지름은 그 두 중심을 잇는 간선을 지남
트리의 모든 지름이 공유하는 정점들의 집합은 항상 하나의 연결된 경로를 이룸

평면그래프

평면 그래프에서 $v - e + f = 1 + c$ (v:정점, e: 간선, f: 면, c: 연결요소 수)
연결된 평면 그래프에서 $v \geq 3$인 중복 간선없는 단순 평면 그래프는 $e \leq 3v-6$을 만족
$2e =$각 면에서 변의 개수의 합$\geq 3f$이므로 $v-e+f = 2$와 연립하면 $e \leq 3v-6$
(연결된 평면 그래프에서) 이분그래프라면 사이클의 최소 길이가 4이므로 $2e \geq 4f, e \leq 2v-4$

매칭

Gallai's Identity: $|IS_{max}| + |VC_{min}| = |V|$
IS_{max}계산은 일반그래프에선 NP-hard
Kőnig's theorem: 이분그래프에서 최대매칭=$|VC_{min}|$
$IS_{max,L} = L - VC_{min,L}, IS_{max,R} = R - VC_{min,R}$
Dilworth's theorem: 부분 순서 집합에서 최대 반사슬의 크기는 사슬 분할의 최소 개수와 같음
antichain은 v_{in}이 $IS_{max,L}$이고 && v_{out}이 $IS_{max,R}$인 v들을 모으면 됨
clique는 complement graph에서의 Independent set과 같음
hall's marriage theorem: 이분그래프 $G = (L + R, E)$에서 L의 부분집합 S에 대해 이와 연결된 R의 부분집합을 $N(S)$라 할 때, 이 이분그래프에 perfect matching이 존재할 필요충분조건은 모든 S에 대하여 $|S| \leq |N(S)|$
-> k-정규 이분 그래프는 항상 perfect matching을 가짐

Degree Bound

무방향 그래프에서 $\sum_{(u,v)\in E} \min(\deg(u), \deg(v)) = O(m\sqrt{m})$

5 Math

5.1 Linear Sieve

```
vector<int> getLpf(int n) {
  vector<int> primes, lpf(n + 1); // least prime factor //
  O(N)
  for (int i = 2; i <= n; i++) {
    if (!lpf[i]) {
      lpf[i] = i;
      primes.push_back(i);
    }
    for (auto p : primes) {
      if (i > n / p) break;
      lpf[i * p] = p;
      if (i % p == 0) break;
    }
  }
  return lpf;
}
```

5.2 Multiplicative Function

```
vector<ll> getMulFunc(int n) { // O(N)
  vector<ll> func(n + 1);
  func[1] = 1;
  vector<int> primes, lpf(n + 1), lpe(n + 1); // least prime
  factor, least prime exponent
  for (ll i = 2; i <= n; i++) {
    if (!lpf[i]) {
      lpf[i] = i;
      lpe[i] = 1;
      primes.push_back(i);
      // func[i] = ? // func(p) // i는 소수
      // ex) mu[i] = -1;
    }
    for (auto p : primes) {
      if (i * p > n) break;
      lpf[i * p] = p;
      if (i % p == 0) {
        lpe[i * p] = lpe[i] + 1;
        // func[i * p] = ? // func(i' * p * p) // 소인수 p의
        지수 1 증가
        // ex) mu[i * p] = 0;
        break;
      }
      else {
        lpe[i * p] = 1;
        func[i * p] = func[i] * func[p]; // gcd(i, p) = 1
      }
    }
  }
  return func;
}
```

5.3 Burnside's Lemma

```
ll bracelet(int c, int n) { // 색깔 c종류 염주순열 개수
  vector<ll> powc(n + 1, 1);
  for (int i = 1; i <= n; i++) powc[i] = powc[i - 1] * c;
  ll a = powc[n / 2 + 1];
  ll b = powc[n / 2];
  ll ans = (n & 1) ? n * a : n / 2 * (a + b);
  for (int i = 0; i < n; i++) ans += powc[__gcd(i, n)];
  return ans / (2 * n);
}
```

5.4 Freivalds' Algorithm

```
template <typename T>
bool freivalds(const vector<vector<T>> &a, const
vector<vector<T>> &b, const vector<vector<T>> &c) { // O(K
N^2)
  int n = a.size();
  vector<unsigned long long> Bx(n), x(n);
  static Random<int> rand(0, 1);
  constexpr int K = 30;
  for (int it = K; it--;) {
    for (auto &e : x) e = rand();
    for (int i = 0; i < n; i++) {
      Bx[i] = 0;
      for (int j = 0; j < n; j++) Bx[i] += b[i][j] * x[j];
    }
    for (int i = 0; i < n; i++) {
      unsigned long long ABx_i = 0, Cx_i = 0;
      for (int j = 0; j < n; j++) {
        ABx_i += a[i][j] * Bx[j];
        Cx_i += c[i][j] * x[j];
      }
      if (ABx_i != Cx_i) return false;
    }
  }
  return true;
}
// n*n행렬 A, B, C에 대해 AB==C(mod 2^64)확인
```

5.5 egcd

```
tuple<ll, ll, ll> egcd(ll a, ll b) { // ax + by = gcd(a, b)
  if (b == 0) return {1, 0, a};
  auto [x, y, g] = egcd(b, a % b);
  return {y, x - (a / b) * y, g};
} // x<=b/g, y<=a/g
ll modInv(ll a, ll b) {
  auto [x, y, g] = egcd(a, b);
  return g != 1 ? -1 : (x + b) % b;
} // modInv(n, MOD)
vector<ll> inv(n + 1, 1);
for (int i = 2; i <= n; ++i) inv[i] = inv[p % i] * (p - p / i)
% p;
```

5.6 FFT

```
// 4 * n * max_{k}(sum|a[i]*b[k-i]|) << 2^53 : double
namespace Poly { // FFT
template <typename real_t>
void fft(vector<complex<real_t>> &a, bool inv) {
  using cpx = complex<real_t>;
  int n = a.size(), L = __lg(n);
  static vector<complex<long double>> R(2, 1);
  static vector<cpx> rt(2, 1);
  for (static int k = 2; k < n; k *= 2) {
    R.resize(n); rt.resize(n);
    auto x = polar(1.0L, acos(-1.0L) / k);
    for (int i = k; i < 2 * k; i++) rt[i] = R[i] = i&1 ?
    R[i/2] * x : R[i/2];
```

```
  }
  vector<int> rev(n);
  for (int i = 0; i < n; i++) rev[i] = (rev[i / 2] | (i & 1)
  << L) / 2;
  for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i],
  a[rev[i]]);
  for (int k = 1; k < n; k *= 2) {
    for (int i = 0; i < n; i += 2 * k) {
      for (int j = 0; j < k; j++) {
        auto x = (real_t *)&rt[j+k], y = (real_t *)&a[i+j+k];
        cpx z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
        // cpx z = rt[j + k] * a[i + j + k];
        a[i + j + k] = a[i + j] - z;
        a[i + j] += z;
      }
    }
  }
  if (inv) {
    reverse(a.begin() + 1, a.end());
    for (auto &e : a) e /= real_t(a.size());
  }
}
template <typename real_t, typename T>
vector<ll> multiply(const vector<T> &A, const vector<T> &B) {
  assert(!A.empty() && !B.empty());
  using cpx = complex<real_t>;
  int need = A.size() + B.size() - 1;
  int n = 1;
  while (n < need) n <<= 1;
  vector<cpx> in(n), out(n);
  for (int i = 0; i < A.size(); i++) in[i] = A[i];
  for (int i = 0; i < B.size(); i++) in[i].imag(B[i]);
  fft(in, false);
  for (auto &x : in) x *= x;
  for (int i = 0; i < n; i++) out[i] = in[-i & (n - 1)] -
  conj(in[i]);
  fft(out, false);
  vector<ll> res(need);
  for (int i = 0; i < res.size(); i++) res[i] =
  llround(imag(out[i]) / (4 * n));
  return res;
}
}

namespace Poly { // 정확도 높은 FFT
template <typename real_t>
void fftPrecisely(vector<complex<real_t>> &a, bool inv) {
  using cpx = complex<real_t>;
  int n = a.size(), L = __lg(n);
  vector<int> rev(n);
  for (int i = 0; i < n; i++) rev[i] = (rev[i >> 1] | ((i & 1)
  << L)) >> 1;
  for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i],
  a[rev[i]]);
  vector<cpx> roots(n / 2);
  real_t ang = 2 * acos(real_t(-1)) / n * (inv ? -1 : 1);
  for (int i = 0; i < n / 2; i++) roots[i] = cpx(cos(ang * i),
  sin(ang * i));
  for (int m = 2; m <= n; m <<= 1) {
    int step = n / m;
    for (int i = 0; i < n; i += m) {
      for (int j = 0; j < m / 2; j++) {
        cpx even = a[i + j];
        cpx odd = roots[step * j] * a[i + j + m / 2];
        a[i + j] = even + odd;
        a[i + j + m / 2] = even - odd;
      }
    }
  }
  if (inv) for (auto &e : a) e /= cpx(n);
}
static constexpr int SPLIT_BIT = 15;
static constexpr ll SPLIT_BASE = 1LL << SPLIT_BIT;
static constexpr ll SPLIT_BASE_SQ = 1LL << SPLIT_BIT * 2;
static constexpr int SPLIT_MASK = (1 << SPLIT_BIT) - 1;
template <typename real_t=double, typename T>
vector<ll> multiplyPrecisely(const vector<T> &A, const
vector<T> &B) {
  assert(!A.empty() && !B.empty());
  using cpx = complex<real_t>;
  int need = A.size() + B.size() - 1;
  int n = 1;
  while (n < need) n <<= 1;
  vector<cpx> a(n), b(n), c1(n), c2(n);
  // for (int i = 0; i < A.size(); i++) a[i] = cpx(A[i] >>
  SPLIT_BIT, A[i] & SPLIT_MASK); // A[i],B[i]>0
  // for (int i = 0; i < B.size(); i++) b[i] = cpx(B[i] >>
  SPLIT_BIT, B[i] & SPLIT_MASK);
  for (int i = 0; i < A.size(); i++) a[i] = cpx(A[i] /
  SPLIT_BASE, A[i] % SPLIT_BASE);
  for (int i = 0; i < B.size(); i++) b[i] = cpx(B[i] /
  SPLIT_BASE, B[i] % SPLIT_BASE);
  fftPrecisely(a, false);
  fftPrecisely(b, false);
  for (int i = 0; i < n; i++) {
    int j = (i ? (n - i) : i);
    cpx ans1 = (a[i] + conj(a[j])) * cpx(0.5, 0);
    cpx ans2 = (a[i] - conj(a[j])) * cpx(0, -0.5);
    cpx ans3 = (b[i] + conj(b[j])) * cpx(0.5, 0);
    cpx ans4 = (b[i] - conj(b[j])) * cpx(0, -0.5);
    c1[i] = (ans1 * ans3) + (ans1 * ans4) * cpx(0, 1);
    c2[i] = (ans2 * ans3) + (ans2 * ans4) * cpx(0, 1);
  }
  fftPrecisely(c1, true);
  fftPrecisely(c2, true);
  vector<ll> res(need);
  for (int i = 0; i < res.size(); i++) {
    ll av = llround(c1[i].real());
    ll bv = llround(c1[i].imag()) + llround(c2[i].real());
    ll cv = llround(c2[i].imag());
    // res[i] = (av << (SPLIT_BIT * 2)) + (bv << SPLIT_BIT) +
    cv; // av,bv,cv>0
    res[i] = av * SPLIT_BASE_SQ + bv * SPLIT_BASE + cv;
  }
  return res;
}
}
// multiply<double_t>(f, g) : fft 다항식 곱셈 O(N logN)
// multiplyPrecisely<double_t>(f, g) : 정확도 높은 fft 다항식
곱셈 O(N logN)
```

5.7 FFT Division

```
namespace Poly { // $O(K^2 \log{N})$ 키타마사
template <typename T>
vector<ll> divideModNaive(ll n, const vector<T> &g, const ll
MOD) { // f(x)=x^n에 대해 f % g 계산 // g 최고차항 계수는 1 //
g차수 k일 때 O(K^2 logN)
  const int k = g.size() - 1;
  assert(g[k] == 1);
  vector<ll> res(k), xn(k);
  res[0] = 1;
  if (k == 1) {
    xn[0] = (MOD - g[0]) % MOD; // x === -g[0] (mod x + g[0])
    if (xn[0] < 0) xn[0] += MOD;
  }
  else xn[1] = 1; // x
  auto add = [&MOD](ll &a, ll b) { a = (a + b) % MOD; };
  auto sub = [&MOD](ll &a, ll b) { a = (a - b) % MOD; if (a <
  0) a += MOD; };
  auto mul = [&](vector<ll> &a, const vector<ll> &b) {
    vector<ll> scratch(2 * k - 1);
    for (int i = 0; i < k; i++) if (a[i]) for (int j = 0; j <
    k; ++j) add(scratch[i + j], a[i] * b[j]); // scratch = a *
    b
    for (int i = 2 * k - 2; i >= k; i--) if (scratch[i]) for
    (int j = 1; j <= k; ++j) sub(scratch[i - j], scratch[i] *
    g[k - j]); // scratch %= g
    for (int i = 0; i < k; i++) a[i] = scratch[i];
  };
  while (n) {
    if (n & 1) mul(res, xn);
    mul(xn, xn);
    n >>= 1;
  }
  return res;
}
template <typename T>
ll kitamasaNaive(ll n, const vector<T> &a, const vector<T>
&coef, const ll MOD) { // a는 초항{a1, a2, ..., ak}, c는
계수{c1, c2, ..., ck} // A_n = (C_1 x A_n-1 + C_2 x A_n-2 +
... + C_k x A_n-k) mod M일 때 A_n 계산
  assert(n >= 1 && !coef.empty() && a.size() == coef.size());
  if (n <= a.size()) return (a[n - 1] % MOD + MOD) % MOD;
```

```
  vector<ll> f(coef.size() + 1, 1); // x^k - c_1 x^(k-1) - c_2
  x^(k-2) - ... - c_k x^0
  for (int i = 0; i < coef.size(); i++) f[coef.size() - 1 - i]
  = (MOD - coef[i]) % MOD;
  auto d = divideModNaive(n - 1, f, MOD); // x^(n-1)을 {x^k -
  c_1 x^(k-1) - c_2 x^(k-2) - ... - c_k x^0}로 나눈 나머지와
  같음
  ll res = 0; // an = a1 * d1 + a2 * d2 + ... + ak * dk
  for (int i = 0; i < d.size(); i++) res = (res + a[i] * d[i])
  % MOD;
  return res;
}
}

namespace Poly { // NTT 다항식 나눗셈, 키타마사 // $O(K
\log{K} \log{N})$ 키타마사
template <ll p, ll primitiveRoot, typename T>
vector<ll> invertMod(const vector<T> &f, int deg) { // f(x)^-1
mod x^deg 계산 // 계수는 mod p로 계산
  assert(!f.empty() && f[0] % p != 0); // f(x)의 상수항이
  0이면 역원이 존재하지 않음 // 참고로 divideMod(f, g)에서 g의
  계수를 뒤집은 게 invertMod의 f이므로 f[0]!=0이란 건
  divideMod에서 g의 최고차항의 계수가 0이 아니란 것과 동치임
  vector<ll> res = {mint<p>(f[0] % p).inv().value}; // f(x)^-1
  mod x^1
  for (int curDeg = 1; curDeg < deg; curDeg <<= 1) {
    int nxtDeg = curDeg << 1;

    // prod = f(x) * res(x)
    vector<ll> prod = multiplyMod<p, primitiveRoot>(res,
    vector<ll>(f.begin(), f.begin() + min(int(f.size()),
    nxtDeg)));
    prod.resize(nxtDeg);

    // prod = 2 - f(x) * res(x)
    if ((prod[0] -= 2) < 0) prod[0] += p;
    for (int i = 0; i < prod.size(); i++) if (prod[i] != 0)
    prod[i] = p - prod[i];

    // 새로운 res(x) = res(x) * (2 - f(x) * res(x))
    res = multiplyMod<p, primitiveRoot>(res, prod);
    res.resize(nxtDeg);
  }

  res.resize(deg);
  return res;
}

template <ll p, ll primitiveRoot, typename T>
pair<vector<ll>, vector<ll>> divideMod(const vector<T> &f,
const vector<T> &g) {
  int n = f.size(), m = g.size();
  if (n < m) {
    vector<ll> res(n);
    for (int i = 0; i < n; i++) {
      res[i] = f[i] % p;
      if (res[i] < 0) res[i] += p;
    }
    return {vector<ll>{0}, res};
  }

  vector<ll> F(f.rbegin(), f.rend());
  vector<ll> G(g.rbegin(), g.rend());

  vector<ll> q = multiplyMod<p, primitiveRoot>(F, invertMod<p,
  primitiveRoot>(G, n - m + 1));
  q.resize(n - m + 1);
  reverse(q.begin(), q.end());

  // r(x) = f(x) - q(x) * g(x)
  vector<ll> r = multiplyMod<p, primitiveRoot>(q, g);
  r.resize(m - 1);

  for (int i = 0; i < m - 1; i++) {
    r[i] = (f[i] - r[i]) % p;
    if (r[i] < 0) r[i] += p;
  }

  return {q, r};
}

template <ll p, ll primitiveRoot, typename T>
vector<ll> remainderMod(const vector<T> &f, const vector<T>
&g) {
  return divideMod<p, primitiveRoot>(f, g).second;
}

template <ll p, ll primitiveRoot, typename T>
vector<ll> divideMod(ll n, const vector<T> &g) { // f(x)=x^n에
대해 f % g 계산 // g차수 k일 때 O(K logK logN)
  vector<ll> res = {1};
  vector<ll> xn = {0, 1};
  while (n) {
    if (n & 1) res = remainderMod<p,
    primitiveRoot>(multiplyMod<p, primitiveRoot>(res, xn), g);
    xn = remainderMod<p, primitiveRoot>(multiplyMod<p,
    primitiveRoot>(xn, xn), g);
    n >>= 1;
  }
  return res;
}
template <ll p, ll primitiveRoot, typename T> // p = a * 2^b +
1 꼴 소수
ll kitamasaNTT(ll n, const vector<T> &a, const vector<T>
&coef) { // a는 초항{a1, a2, ..., ak}, c는 계수{c1, c2, ...,
ck} // A_n = (C_1 x A_n-1 + C_2 x A_n-2 + ... + C_k x A_n-k)
mod P일 때 A_n 계산
  assert(n >= 1 && !coef.empty() && a.size() == coef.size());
  if (n <= a.size()) return a[n - 1];
  vector<ll> f(coef.size() + 1, 1); // x^k - c_1 x^(k-1) - c_2
  x^(k-2) - ... - c_k x^0
  for (int i = 0; i < coef.size(); i++) f[coef.size() - 1 - i]
  = -coef[i];
  auto d = divideMod<p, primitiveRoot>(n - 1, f); // x^(n-1)을
  {x^k - c_1 x^(k-1) - c_2 x^(k-2) - ... - c_k x^0}로 나눈
  나머지
  ll res = 0; // an = a1 * d1 + a2 * d2 + ... + ak * dk
  for (int i = 0; i < d.size(); i++) res = (res + a[i] * d[i])
  % p;
  return res;
}
}
/*
multiplyMod<p, primitiveRoot>(f, g) : ntt 다항식 곱셈 O(N
logN)
invertMod<p, primitiveRoot>(f, deg) : $f(x)^{-1}
\pmod{x^\text{deg}}$ 계산 | $O(N \log{N})$ |
divideMod<p, primitiveRoot>(f, g)  {$f(x) \text{ / } g(x)$,
$f(x) \text{ \% } g(x)$} 계산 | $O(N \log{N})$ |
divideMod<p, primitiveRoot>(n, g)  $x^n \pmod{g(x)}$ 계산 |
$O(K \log{K} \log{N})$ ($K$는 $g(x)$의 최고차항의 차수)
divideModNaive(n, g, MOD)  $x^n \pmod{g(x)}$ 계산 | $O(K^2
\log{N})$ ($K$는 $g(x)$의 최고차항의 차수)
kitamasaNTT<p, primitiveRoot>(n, a, coef)  초항 ${a_1, ...,
a_k}$과 점화식 $a_i = c_1 a_{i-1} + ... + c_k a_{i-k}$에 대해
$a_n \mod{p}$ 계산 | $O(K \log{K} \log{N})$ ($K$는 $g(x)$의
최고차항의 차수)
kitamasaNaive(n, a, coef, MOD)  초항 ${a_1, ..., a_k}$과
점화식 $a_i = c_1 a_{i-1} + ... + c_k a_{i-k}$에 대해 $a_n
\mod{\text{MOD}}$ 계산 | $O(K^2 \log{N})$ ($K$는 $g(x)$의
최고차항의 차수)
*/
```

5.8 Multipoint Evaluation

```
namespace Poly { // multipoint evaluation
  template <ll p, ll primitiveRoot, typename T>
  vector<ll> multiEval(const vector<T> &f, const vector<T>
  &qs) { // O(NlogN + qlog^2q)
    if (qs.empty()) return {};
    assert(!f.empty());
    vector<vector<ll>> tree(4 * qs.size());
    auto build = [&](auto &&build, int node, int s, int e) {
      if (s == e) {
        tree[node] = {(p - qs[s] % p) % p, 1};
        return;
      }
      int m = s + e >> 1;
      build(build, node << 1, s, m);
      build(build, node << 1 | 1, m + 1, e);
      tree[node] = multiplyMod<p, primitiveRoot>(tree[node <<
      1], tree[node << 1 | 1]);
    };
    build(build, 1, 0, qs.size() - 1); // O(qlog^2q)
```

```
    vector<ll> res(qs.size());
    auto eval = [&](auto &&eval, vector<ll> poly, int node,
    int s, int e) {
      poly = remainderMod<p, primitiveRoot>(poly, tree[node]);
      if (s == e) {
        assert(!poly.empty());
        res[s] = poly[0];
        return;
      }
      int m = s + e >> 1;
      eval(eval, poly, node << 1, s, m);
      eval(eval, poly, node << 1 | 1, m + 1, e);
    };
    eval(eval, f, 1, 0, qs.size() - 1);
    return res;
  }
}
```

5.9 NTT

```
namespace Poly { // NTT
template <ll p> ll binpow(ll a, ll n) { //
(MOD-1)^2<=INT64_MAX
  ll res = 1;
  for (; n; n >>= 1) {
    if (n & 1) res = res * a % p;
    a = a * a % p;
  }
  return res;
}
template <ll p, ll primitiveRoot>
void ntt(vector<ll> &a, bool inv) {
  int n = a.size(), L = __lg(n);
  static vector<ll> rt(2, 1);
  for (static int k = 2, s = 2; k < n; k <<= 1, ++s) {
    rt.resize(n);
    ll z[2] = {1, binpow<p>(primitiveRoot, p >> s)};
    for (int i = k; i < 2 * k; i++) rt[i] = rt[i >> 1] * z[i &
    1] % p;
  }
  vector<int> rev(n);
  for (int i = 0; i < n; i++) rev[i] = (rev[i >> 1] | ((i & 1)
  << L)) >> 1;
  for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i],
  a[rev[i]]);
  for (int k = 1; k < n; k <<= 1) {
    for (int i = 0; i < n; i += 2 * k) {
      for (int j = 0; j < k; j++) {
        ll u = a[i + j], v = a[i + j + k] * rt[j + k] % p;
        a[i + j] = u + v < p ? u + v : u + v - p;
        a[i + j + k] = u - v >= 0 ? u - v : u - v + p;
      }
    }
  }
  if (inv) {
    reverse(a.begin() + 1, a.end());
    ll invN = binpow<p>(n, p - 2);
    for (auto &e : a) e = e * invN % p;
  }
}
template <ll p, ll primitiveRoot, typename T>
vector<ll> multiplyMod(const vector<T> &A, const vector<T> &B)
{
  assert(!A.empty() && !B.empty());
  int need = A.size() + B.size() - 1;
  int n = 1;
  while (n < need) n <<= 1;
  assert(n <= ((p - 1) & -(p - 1))); // assert(n <= 2^b)
  vector<ll> a(n), b(n);
  for (int i = 0; i < A.size(); i++) {
    a[i] = A[i] % p;
    if (a[i] < 0) a[i] += p;
  }
  for (int i = 0; i < B.size(); i++) {
    b[i] = B[i] % p;
    if (b[i] < 0) b[i] += p;
  }
  ntt<p, primitiveRoot>(a, false);
  ntt<p, primitiveRoot>(b, false);
  for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % p;
  ntt<p, primitiveRoot>(a, true);
  a.resize(need);
  return a;
}
// | p = a*2^b+1   | a   | b  | 2^b      | g |
// | ------------- | --- | -- | -------- | - |
// | 104'857'601   | 25  | 22 | 4194304  | 3 |
// | 167'772'161   | 5   | 25 | 33554432 | 3 |
// | 469'762'049   | 7   | 26 | 67108864 | 3 |
// | 998'244'353   | 119 | 23 | 8388608  | 3 |
// | 1'004'535'809 | 479 | 21 | 2097152  | 3 |
// | 1'012'924'417 | 483 | 21 | 2097152  | 5 |
// | 2'281'701'377 | 17  | 27 | 134217728| 3 |
// | 2'483'027'969 | 37  | 26 | 67108864 | 3 |
}
```

5.10 CRT

```
pll merge(const pll &a, const pll &b) {
  auto [p1, m1] = a;
  auto [p2, m2] = b;
  ll g = __gcd(p1, p2);
  if ((m2 - m1) % g) return {-1, -1};
  ll x = (m2 - m1) / g * modInverse(p1 / g, p2 / g) % (p2 /
  g);
  return {p1 / g * p2, p1 * x + m1};
}
pll crt(const vector<pll> &rems) { // rems 원소는 {p, n % p}꼴
// O(N logP) (P:= p1*p2*...*pn)
  pll res(1, 0);
  for (auto &p : rems) {
    res = merge(res, p);
    if (res.first == -1) return {-1, -1};
  }
  if (res.second < 0) res.second += res.first;
  return res;
} // crt().first = -1이면 연립합동식에 해 존재
```

5.11 Miller-Rabin

```
inline ll multiply(ll a, ll b, ll mod) { return __int128(a) *
b % mod; }
ll power(ll a, ll n, ll mod) { //a ^ n % mod
  ll res = 1;
  while (n) {
    if (n & 1) res = multiply(res, a, mod);
    a = multiply(a, a, mod);
    n >>= 1;
  }
  return res;
}
bool isPrime(ll n) { // 밀러-라빈, O(7log^3N)
  if (n <= 1) return false;
  ll d = n - 1, r = 0;
  while (~d & 1) d >>= 1, ++r;
  auto check = [&](ll a) {
    ll remain = power(a, d, n);
    if (remain == 1 || remain == n - 1) return true;
    for (int i = 1; i < r; i++) {
      remain = multiply(remain, remain, n);
      if (remain == n - 1) return true;
    }
    return false;
  };
  vector<ll> nums = {2,325, 9375, 28178, 450775, 9780504,
  1795265022}; // ull
  // int 범위는 {2, 7, 61}
  for (ll a : nums) if (a % n && !check(a)) return false;
  return true;
}
```

5.12 Pollard's rho

```
namespace PollardRho {
  ll getPrime(ll n) {
    if (~n & 1) return 2;
    if (isPrime(n)) return n;
    ll a, b, c, g;
    auto f = [&c, &n](ll x) { return (multiply(x, x, n) + c) %
    n; };
    do {
      a = b = rand() % (n - 2) + 2;
      c = rand() % 10 + 1;
      do {
        a = f(a);
        b = f(f(b));
        g = __gcd(abs(a - b), n);
      } while (g == 1);
    } while (g == n);
    return getPrime(g);
```

```
  }
  vector<pair<ll, ll>> factorize(ll n) { // O(\sqrt[4]N)
    assert(n > 1);
    vector<pair<ll, ll>> primes;
    while (n > 1) {
      ll prime = getPrime(n), cnt = 0;
      while (n % prime == 0) n /= prime, ++cnt;
      primes.push_back({prime, cnt});
    }
    sort(primes.begin(), primes.end());
    return primes;
  }
  vector<ll> getAllDivisors(ll n) {
    vector<ll> res = {1};
    if (n == 1) return res;
    for (auto [p, e] : factorize(n)) {
      int sz = res.size();
      ll curP = p;
      for (int _ = e; _--;) {
        for (int i = 0; i < sz; i++) res.push_back(res[i] *
        curP);
        curP *= p;
      }
    }
    assert(false, "정렬 필요한지 확인");
    // sort(res.begin(), res.end());
    return res;
  }
}
```

5.13 Tonelli-Shanks

```
inline ll multiply(ll a, ll b, ll mod) { return __int128(a) *
b % mod; }
ll binpow(ll a, ll n, ll mod) { //a ^ n % mod
  ll res = 1;
  while (n) {
    if (n & 1) res = multiply(res, a, mod);
    a = multiply(a, a, mod);
    n >>= 1;
  }
  return res;
}
// p=4k+3이고 n이 이차잉여라면 이산제곱근은 n^((p+1)/4)
ll tonelliShanks(ll n, ll p) { // O(log^2(p)) // x^2 === n
(mod p) // 해가 없다면 -1 리턴
  assert (n >= 0 && n < p);
  if (!n) return 0;
  if (p == 2) return n;
  if (binpow(n, (p - 1) / 2, p) != 1) return -1;
  ll Q = p - 1, S = 0;
  while (~Q & 1) {
    Q >>= 1;
    ++S;
  }
  ll z = 2;
  while (binpow(z, (p - 1) / 2, p) != p - 1) ++z;
  ll M = S;
  ll c = binpow(z, Q, p);
  ll t = binpow(n, Q, p);
  ll R = binpow(n, (Q + 1) / 2, p);
  while (t != 1) {
    ll i = [&]() {
      ll tmp = t;
      for (int i = 1; i < M; i++) {
        tmp = multiply(tmp, tmp, p);
        if (tmp == 1) return i;
      }
    }();
    ll b = binpow(c, binpow(2, M - i - 1, p - 1), p); // c ^
    (2 ^ (M - i - 1)), phi(p) = p - 1
    M = i;
    c = multiply(b, b, p);
    t = multiply(t, c, p);
    R = multiply(R, b, p);
  }
  return R; // (p - R) % p도 해가 되긴 함
}
```

5.14 Cornacchia

```
ll isqrt(ll n) {
  assert(n >= 0);
  if (n == 0) return 0LL;
  ll x = sqrtl(n);
  while ((x + 1) <= n / (x + 1)) ++x;
  while (x > n / x) --x;
  return x;
}
pll cornacchia(ll m, ll d) { // x^2 + d y^2 = m // m은 소수,
d>0 // 해 없으면 {-1, -1} // O(log^2 m)
  assert(m >= 2 && d > 0);
  ll r0 = tonelliShanks((m - d % m) % m, m);
  if (r0 == -1) return {-1, -1};
  ll prv = m, cur = r0;
  while ((__int128)cur * cur >= m) {
    ll nxt = prv % cur;
    prv = cur;
    cur = nxt;
  }
  ll x = cur;
  ll rem = m - x * x;
  if (rem < 0 || rem % d != 0) return {-1, -1};
  ll y2 = rem / d;
  ll y = isqrt(y2);
  if ((__int128)y * y != y2) return {-1, -1};
  return {x, y};
}
```

5.15 Floorsum

```
ll f(int p, int q, int n) { // p/q + 2p/q + ... + np/q //
O(log(min(p, q)))
  assert(q != 0);
  if (p == 0) return 0;
  if (p >= q) return ll(p / q) * n * (n + 1) / 2 + f(p % q, q,
  n);
  return ll(n) * p / q * n + n / q - f(q, p, ll(p) * n / q);
}
ll floorModSum(int p, int q, int n) { // p % q + 2p % q + ...
+ np % q // O(log(min(p, q)))
  int g = __gcd(p, q);
  int pg = p / g, qg = q / g;
  return ll(p) * n * (n + 1) / 2 - q * f(pg, qg, n);
}
```

5.16 Xudyh's Sieve

```
template <typename T, T MOD>
class Xudyh {
  vector<T> pfsF; // precomputation
  map<ll, T> mp; // memoization
  T positiveMod(T x) {
    return (x % MOD + MOD) % MOD;
  }
  T pfs_fg(ll x) { // (f * g)(x) =
    // x %= MOD;
    // return positiveMod();
  };
  T pfs_g(ll x) { // g(x) =
    // x %= MOD;
    // return positiveMod();
  };
  T pfs_f(ll x) { // f(x) =
    if (x < pfsF.size()) return pfsF[x]; // precomputation
    if (mp.count(x)) return mp[x];
    auto &res = mp[x];
    res = pfs_fg(x); // S_f(N) = (  S_f*g(N) - sum {d = 2~N}
    g(d)S_f(N/d)  ) / g(1)
    for (ll i = 2, j; i <= x; i = j + 1) {
      j = x / (x / i);
      res -= pfs_f(x / i) * (pfs_g(j) - pfs_g(i - 1) + MOD) %
      MOD;
      res %= MOD;
    }
    res = (res + MOD) % MOD;
    res = res * modInv(g(1), MOD) % MOD; // res /= g(1)
    return res;
  }
public:
  Xudyh(ll maxN) {
    int preComputeSize = pow(2.0L * maxN, 2.0L / 3);
    pfsF = getMulFunc(preComputeSize);
    for (int i = 1; i < pfsF.size(); i++) pfsF[i] = (pfsF[i] +
    pfsF[i - 1]) % MOD;
  }
  T getSum(ll i, ll j) { // O(N^{2/3}) // precomputation 안 할
  경우 O(N^{3/4}) // precomputation K까지했을 때 O(K +
  \dfrac{2N}{\sqrt{K}})
    return positiveMod(pfs_f(j) - pfs_f(i - 1));
```

```
  }
};
// g(1)S_f(N) = S_{f*g}(N) - sum_{d=2}^{N} g(d) S_f(N/d))$
// $S_f(N) = (S_{f*g}(N) - sum_{d=2}^{N} g(d) S_f(N/d)) ) / g(1)
```

5.17 Lagrange Interpolation

```
ll pm(ll n, ll mod) {
  return (n % mod + mod) % mod;
}
// f(0), ..., f(n) 알고 있을 때 n차 다항식 f(x)의 x좌표에서 함수값 계산
ll lagrangeInterpolation(vector<ll> y, ll x, ll mod) { // O(N \log{MOD})
  assert(!y.empty());
  for (auto &e : y) e = pm(e, mod);
  x = pm(x, mod);
  int n = y.size() - 1;
  if (x <= n) return pm(y[x], mod);
  ll k = 1; // PI_{j!=i} (x - x_j) / (x_i - x_j) 저장
  for (int j = 1; j < y.size(); j++) { // i = 0 // j != i
    k = k * pm(x - j, mod) % mod; // * (x - x_j)
    k = k * modInv(pm(-j, mod), mod) % mod; // / (x_i - x_j)
  }
  ll res = 0;
  for (int i = 0; i <= n; i++) {
    res = (res + y[i] * k) % mod;
    if (i < n) {
      k = k * pm(x - i, mod) % mod * modInv(pm(x - (i + 1), mod), mod) % mod;
      k = k * pm(i - n, mod) % mod * modInv(pm(i + 1, mod), mod) % mod;
    }
  }
  return res;
}
// (x_0, y_0), ..., (x_n, y_n)을 지나는 n차함수:
// y=sum_{i=0}^{n} [ y_i prod_{j \neq i}(x-x_j) / prod_{j \neq i}(x_i-x_j) ]

ll powerSum(ll n, int k, ll mod) { // O(K log{MOD}) + O(K logK) // (1^k + 2^k + ... + n^k) mod M
  vector<ll> y(k + 2, 0);
  for (int x = 1; x <= k + 1; x++) y[x] = (y[x - 1] + power(x, k, mod)) % mod;
  return n < y.size() ? y[n] : lagrangeInterpolation(y, n, mod);
}
```

5.18 Moebius Inversion

$\tau(n)$: 양의 약수의 개수
$\sigma(n)$: 양의 약수의 합
$\phi(n)$: n 이하의 자연수 중 n과 서로소인 수의 개수 ($\phi(1) = 1$)
$\mu(n)$: 뫼비우스 함수
$\varepsilon(n)$: $[n = 1]$ (Iverson bracket)
$\mathbf{1}(n) = 1$, $\mathrm{Id}(n) = n$

Dirichlet Convolution

$$\tau = \mathbf{1} * \mathbf{1}, \quad \sigma = \mathrm{Id} * \mathbf{1}, \quad \mathrm{Id} = \phi * \mathbf{1}, \quad \mathbf{1} = \varepsilon * \mathbf{1}, \quad \sigma = \phi * \tau$$

Moebius Inversion

$$g(n) = \sum_{d|n} f(d), \quad \forall n \in \mathbb{N} \iff f(n) = \sum_{d|n} g(d)\mu\left(\frac{n}{d}\right), \quad \forall n \in \mathbb{N}$$

즉,

$$g = f * \mathbf{1} \iff f = g * \mu$$

$$\mathbf{1} = \tau * \mu, \quad \mathrm{Id} = \sigma * \mu, \quad \phi = \mathrm{Id} * \mu, \quad \varepsilon = \mathbf{1} * \mu$$

$$\varepsilon = (\mathrm{Id}\,\mu) * \mathrm{Id}$$
$$\mathrm{Id}^{k+1} = (\mathrm{Id}^k\,\phi) * \mathrm{Id}^k$$

GCD Sums

$$\sum_{k=1}^{n}\sum_{d=1}^{\lfloor n/k \rfloor} = \sum_{l=1}^{n}\sum_{d|l} \qquad \left(l = kd,\ d = \frac{l}{k}\right)$$

1.

$$\sum_{i=1}^{n}\sum_{j=1}^{n}[\gcd(i,j)=1] = \sum_{d=1}^{n}\mu(d)\left\lfloor\frac{n}{d}\right\rfloor^2$$

1'.

$$\sum_{i=1}^{n}\sum_{j=1}^{n}[\gcd(i,j)=g] = \sum_{d=1}^{\lfloor n/g \rfloor}\mu(d)\left\lfloor\frac{n}{gd}\right\rfloor^2$$

$$\sum_{i=1}^{n}\sum_{j=1}^{n}[\gcd(i,j)=1] = 2\sum_{i=1}^{n}\phi(i) - 1$$

2.

$$\begin{aligned}
\sum_{i=1}^{n}\sum_{j=1}^{n}\gcd(i,j) &= \sum_{k=1}^{n}k\sum_{i=1}^{n}\sum_{j=1}^{n}[\gcd(i,j)=k] \\
&= \sum_{k=1}^{n}k\sum_{i=1}^{\lfloor n/k \rfloor}\sum_{j=1}^{\lfloor n/k \rfloor}[\gcd(i,j)=1] \\
&= \sum_{k=1}^{n}k\sum_{d=1}^{\lfloor n/k \rfloor}\mu(d)\left\lfloor\frac{\lfloor n/k \rfloor}{d}\right\rfloor^2 \\
&= \sum_{k=1}^{n}k\sum_{d=1}^{\lfloor n/k \rfloor}\mu(d)\left\lfloor\frac{n}{kd}\right\rfloor^2 \\
&= \sum_{l=1}^{n}\left\lfloor\frac{n}{l}\right\rfloor^2\sum_{k|l}k\mu\left(\frac{l}{k}\right) \\
&= \sum_{l=1}^{n}\left\lfloor\frac{n}{l}\right\rfloor^2\phi(l)
\end{aligned}$$

2'.

$$\sum_{i=1}^{n}\sum_{j=1}^{m}\gcd(i,j) = \sum_{l=1}^{\min(n,m)}\left\lfloor\frac{n}{l}\right\rfloor\left\lfloor\frac{m}{l}\right\rfloor\phi(l)$$

2".

$$\sum_{i=1}^{n}\sum_{j=1}^{m}i\gcd(i,j) = \sum_{l=1}^{\min(n,m)}\left\lfloor\frac{m}{l}\right\rfloor\frac{\lfloor n/l \rfloor(\lfloor n/l \rfloor+1)}{2}l\phi(l)$$

2"'.

$$\sum_{i=1}^{n}\sum_{j=1}^{m}ij\gcd(i,j) = \sum_{l=1}^{\min(n,m)}\frac{\lfloor m/l \rfloor(\lfloor m/l \rfloor+1)}{2}\frac{\lfloor n/l \rfloor(\lfloor n/l \rfloor+1)}{2}l^2\phi(l)$$

5.19 Math Theory

$$\sum_{k=0}^{r}\binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$
$$\sum_{i=r}^{n}\binom{i}{r} = \binom{n+1}{r+1}$$
$$\sum_{x=0}^{n}\binom{x+m}{m} = \binom{n+m+1}{m+1}$$
$$\sum_{x=0}^{n}x\binom{x+m}{m} = \sum_{x=0}^{n}x\binom{x+m}{x} = \sum_{x=0}^{n}(m+1)\binom{x+m}{m+1} = (m+1)\binom{n+m+1}{m+2}$$
$$\sum_{x=0}^{n}x^2\binom{x+m}{m} = \sum_{x=1}^{n}(m+1)x\binom{x+m}{m+1} = (m+1)\sum_{x=0}^{n-1}(x+1)\binom{x+m+1}{m+1} = (m+1)(m+2)\binom{n+m+1}{m+3} + (m+1)\binom{n+m+1}{m+2}$$
$$\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k} \qquad (0 \le k \le r \le n)$$

Canonical Coin System

$c_i = \dfrac{r^i - 1}{r - 1}$은 $\forall r \ge 2$에 대해 canonical coin system임. ex) {1, 3, 7, 15}

Jacobsthal's function

n	P_n	Primorial ($P_n\#$)	Jacobsthal $j(P_n\#)$
1	2	2	2
2	3	6	4
3	5	30	6
4	7	210	10
5	11	2,310	14
6	13	30,030	22
7	17	510,510	26
8	19	9,699,690	34
9	23	223,092,870	40
10	29	6,469,693,230	46

$\phi(n) \approx \frac{6}{\pi^2} n \approx 0.6079n$

Maximal Prime Gaps

Upper Bound (N)	Max Gap g_n	Starting Prime p_n
10^3	20	887
10^6	114	492,113
10^9	282	473,266,931
10^{12}	906	999,999,999,763
10^{15}	1096	908,460,772,044,959
10^{18}	1,476	1,425,172,824,437,699,411
2^{64}	1,550	18,361,375,334,787,046,697

Bertrand's Postulate $\forall n > 1, \exists p$ s.t. $n < p < 2n$

moebius

n이하 square-free 수의 개수 $\sum_{i=1}^{n} \mu[i] \left\lfloor \frac{n}{i^2} \right\rfloor$

Frobenius' coin problem

서로소인 두 자연수 a, b에 대해 an + bm꼴로 나타낼 수 없는 최대값은 ab - a - b

즉, $ab - a - b + 1 = (a-1)(b-1)$이상의 자연수는 전부 $an + bm$꼴로 표현 가능

gcd(a, b) = g라면 (a/g - 1)(b/g - 1)g 이상의 g의 배수는 전부 표현 가능

Cycle Count Generating Functions

순열 $\sigma \in S_n$의 cycle 개수를 $c(\sigma)$, 교란순열의 집합을 $\mathcal{D}_n$이라 하면

$$\sum_{\sigma \in S_n} x^{c(\sigma)} = x(x+1)(x+2)\cdots(x+n-1),$$

$$\sum_{\sigma \in \mathcal{D}_n} x^{c(\sigma)} = \sum_{q=0}^{n} (-1)^q \binom{n}{q} x^q \prod_{i=0}^{n-q-1} (x+i).$$

$$\sum_{\sigma \in S_n} (-1)^{c(\sigma)} = \begin{cases} 1 & n = 0, \\ -1 & n = 1, \\ 0 & n \geq 2, \end{cases} \qquad \sum_{\sigma \in \mathcal{D}_n} (-1)^{c(\sigma)} = 1 - n.$$

Non-adjacent Selection

n개 중 r개를 서로 인접하지 않게 고르는 경우의 수 $\binom{n-r+1}{r}$

Pick's Theorem 격자 다각형 넓이 A, 내부 격자점 개수 I, 경계 격자점 개수 B

$$A = I + \frac{B}{2} - 1.$$

$$B = \sum_{\text{edges}} \gcd\left(|x_2 - x_1|,\ |y_2 - y_1|\right).$$

6 String

6.1 Rolling Hash

```
template<ll MOD=1'000'000'007>
struct Hashing {
  vector<ll> h;
  static const ll X;
  static vector<ll> x;
  template <typename T> // T = string or vector<>
  Hashing(const T &st) : h(st.size() + 1) {
    while (x.size() <= st.size()) x.push_back(x.back() * X %
    MOD);
    for (int i = 1; i < h.size(); i++) h[i] = (h[i - 1] * X +
    st[i - 1]) % MOD;
  }
  ll get(int s, int e) const { // 0-based, st[s:e)
    ll res = (h[e] - h[s] * x[e - s]) % MOD; // h는 1-based
    return res >= 0 ? res : res + MOD;
  }
};
template<ll MOD> const ll Hashing<MOD>::X = sqrt(MOD) -
(__TIME__[6] & 15) * 10 - (__TIME__[7] & 15);
template<ll MOD> vector<ll> Hashing<MOD>::x = {1};
// 2D
template<ll MOD=1'000'000'007>
struct Hashing2D {
  vector<vector<ll>> h;
  static const ll X, Y;
  static vector<ll> x, y;
  template <typename T> // T = string or vector<>
  Hashing2D(const vector<T> &v) {
    int n = v.size();
    int m = v[0].size();
    while (x.size() <= n) x.push_back(x.back() * X % MOD);
    while (y.size() <= m) y.push_back(y.back() * Y % MOD);
    h.assign(n + 1, vector<ll>(m + 1));
    for (int i = 1; i <= n; i++) {
      for (int j = 1; j <= m; j++) {
        h[i][j] = v[i - 1][j - 1]
            + h[i - 1][j] * X % MOD
            + h[i][j - 1] * Y % MOD
            - h[i - 1][j - 1] * X % MOD * Y % MOD;
        h[i][j] %= MOD;
        if (h[i][j] < MOD) h[i][j] += MOD;
      }
    }
  }
  ll get(int i1, int i2, int j1, int j2) const { // 0-based,
  v[i1:i2)[j1:j2)
    ll res = h[i2][j2]
         - h[i1][j2] * x[i2 - i1] % MOD
         - h[i2][j1] * y[j2 - j1] % MOD
         + h[i1][j1] * x[i2 - i1] % MOD * y[j2 - j1] % MOD;
    res %= MOD;
    return res >= 0 ? res : res + MOD;
  }
};
template<ll MOD> const ll Hashing2D<MOD>::X = sqrt(MOD) -
(__TIME__[6] & 15) * 10 - (__TIME__[7] & 15);
template<ll MOD> const ll Hashing2D<MOD>::Y = sqrt(MOD) +
(__TIME__[6] & 15) * 10 + (__TIME__[7] & 15);
template<ll MOD> vector<ll> Hashing2D<MOD>::x = {1};
template<ll MOD> vector<ll> Hashing2D<MOD>::y = {1};
```

6.2 KMP

```
template <typename T>
vector<int> getPi(const T &st) { // O(N)
  vector<int> pi(st.size());
  for (int i = 1, j = 0; i < st.size(); i++) {
    while (j > 0 && st[i] != st[j]) j = pi[j - 1];
    if (st[i] == st[j]) pi[i] = ++j;
  }
  return pi;
}
template <typename T>
vector<int> kmp(const T &st, const T &pattern) { // O(N+M)
  int n = st.size(), m = pattern.size();
  auto pi = getPi(pattern);
  vector<int> res;
  for(int i = 0, j = 0; i < n; i++){
    while(j > 0 && st[i] != pattern[j]) j = pi[j - 1];
    if (st[i] == pattern[j]) {
      if(j == m - 1) {
        res.push_back(i - m + 1);
        j = pi[j];
      }
      else ++j;
    }
  }
  return res;
}
// 최소주기: n - pi.back() // ex) st = "abaab", pi.back() = 2
// cyclic string에서 검색: kmp(st + st, pattern)
```

6.3 Aho-Corasick

```
template <int ALPHA = 26, char FIRST = 'a'>
struct Aho {
  struct Node {
```

```
    array<int, ALPHA> mp, go;
    int fail = 0;
    bool end = false;
    Node() {
      mp.fill(-1);
      go.fill(-1);
    }
  };
  vector<Node> tree = {Node()};
  int charToIdx(char ch) const { return ch - FIRST; }
  void insert(const string &st) {
    int cur = 0;
    for (auto ch : st) {
      int i = charToIdx(ch);
      if (!~tree[cur].mp[i]) {
        tree[cur].mp[i] = tree.size();
        tree.emplace_back();
      }
      cur = tree[cur].mp[i];
    }
    tree[cur].end = true;
  }
  void build() { // build O(ALPHA * sum(m_i))
    queue<int> q;
    for (int i = 0; i < ALPHA; i++) {
      int nxt = tree[0].mp[i];
      if (!~nxt) tree[0].go[i] = 0;
      else {
        tree[0].go[i] = nxt;
        q.push(nxt);
      }
    }
    while (!q.empty()) {
      int cur = q.front();
      q.pop();
      tree[cur].end |= tree[tree[cur].fail].end;
      for (int i = 0; i < ALPHA; i++) {
        int nxt = tree[cur].mp[i];
        if (!~nxt) tree[cur].go[i] =
        tree[tree[cur].fail].go[i];
        else {
          tree[cur].go[i] = nxt;
          tree[nxt].fail = tree[tree[cur].fail].go[i];
          q.push(nxt);
        }
      }
    }
  }
  bool findSubstring(const string &st) const { // O(N)
    int cur = 0;
    for (auto ch : st) {
      cur = tree[cur].go[charToIdx(ch)];
      if (tree[cur].end) return true;
    }
    return false;
  }
};
```

6.4 Z

```
template <typename Container>
vector<int> get_z(const Container &st) { // O(N)
  vector<int> z(st.size());
  for (int i = 1, p = 0; i < st.size(); i++) {
    if (i < p + z[p]) z[i] = min(p + z[p] - i, z[i - p]);
    while (i + z[i] < st.size() && st[z[i]] == st[i + z[i]])
    ++z[i];
    if (p + z[p] < i + z[i]) p = i;
  }
  z[0] = st.size();
  return z;
} // z[i] = lcp(st, st[i:])
```

6.5 Manacher

```
template <typename Container>
vector<int> manacher(const Container &orig, typename
Container::value_type dummy) { // O(N)
  Container st(orig.size() << 1 | 1, dummy);
  for (int i = 0; i < orig.size(); i++) st[2 * i + 1] =
  orig[i];
  vector<int> r(st.size());
  for (int i = 1, p = 0; i < st.size(); i++) {
    if (i < p + r[p]) r[i] = min(p + r[p] - i, r[2 * p - i]);
    while (i - r[i] - 1 >= 0 && i + r[i] + 1 < st.size() &&
    st[i - r[i] - 1] == st[i + r[i] + 1]) ++r[i];
    if (p + r[p] < i + r[i]) p = i;
  }
  return r;
} // 중심st[i]인 가장 긴 팰린드롬 길이 : r[2i+1],
중심st[i:i+1] : r[2i+2]
```

6.6 Suffix Array

```
template <typename Container>
auto get_sa(const Container &st) { // O(N logN)
  int n = st.size();
  if (n == 1) return pair{vector{0}, vector{0}};
  assert(n > 0);
  int m = *max_element(st.begin(), st.end());
  vector<int> sa(n), x(n + 1), y(n + 1), cnt(max(n, m) + 1);
  for (int i = 0; i < n; i++) assert(st[i] > 0);
  for (int i = 0; i < n; i++) ++cnt[x[i] = st[i]];
  for (int i = 1; i <= m; i++) cnt[i] += cnt[i - 1];
  for (int i = n - 1; i >= 0; i--) sa[--cnt[x[i]]] = i;
  for (int len = 1, p = 1; p < n; len <<= 1, m = p) {
    int idx = 0;
    for (int i = n - len; i < n; i++) y[idx++] = i;
    for (int i = 0; i < n; i++) if (sa[i] >= len) y[idx++] =
    sa[i] - len;
    fill(cnt.begin(), cnt.begin() + m + 1, 0);
    for (int i = 0; i < n; i++) ++cnt[x[y[i]]];
    for (int i = 1; i <= m; i++) cnt[i] += cnt[i - 1];
    for (int i = n - 1; i >= 0; i--) sa[--cnt[x[y[i]]]] =
    y[i];
    swap(x, y);
    x[sa[0]] = p = 1;
    for (int i = 1; i < n; i++) {
      int a = sa[i - 1], b = sa[i];
      x[b] = y[a] == y[b] && y[a + len] == y[b + len] ? p :
      ++p;
    }
  }
  x.resize(n);
  for (auto &e : x) --e;
  return pair{sa, x}; // rank=x
}
template <typename Container>
vector<int> get_lcp(const Container &st, const vector<int>
&sa, const vector<int> &rank) { // O(N)
  int n = st.size();
  assert(n >= 1);
  vector<int> lcp(n - 1);
  for (int i = 0, h = 0; i < n; ++i, h -= !!h) if (rank[i]) {
    for (int j = sa[rank[i] - 1]; j + h < n && i + h < n &&
    st[j + h] == st[i + h];) ++h;
    lcp[rank[i] - 1] = h;
  }
  return lcp;
} // lcp[i]: sa[i], sa[i + 1]의 최장 공통 접두사 // lcp.size()
= n-1
// auto [sa, rank] = get_sa(st);
// auto lcp = get_lcp(st, sa, rank);
// RMQ<int, min> rmq(lcp);
auto lce = [&](int i, int j) { // lcp(st[i:], st[j:])
  if (i == j) return n - i;
  int idx1 = rank[i];
  int idx2 = rank[j];
  if (idx1 > idx2) swap(idx1, idx2);
  return rmq.qry(idx1, idx2 - 1); // i!=j라서 idx1 <= idx2 - 1
  성립
};
auto cmp = [&](int i, int j, int i2, int j2) { // st[i:j] <
st[i2:j2]
  int len = lce(i, i2);
  if (len >= min(j - i + 1, j2 - i2 + 1)) return j - i < j2 -
  i2;
  return rank[i] < rank[i2];
};
```

6.7 String Theory

Infinite Order

비어 있지 않은 문자열 a, b에 대해 사전순으로 $a^\infty < b^\infty \iff ab < ba$

7 Misc

7.1 Gray Code

```
void grayCode(int n) {
  for (int i = 0; i < (1 << n); ++i) cout << (i ^ (i >> 1)) <<
  "\n";
}
```

7.2 Majority Vote

```
template <typename T>
pair<bool, T> boyerMoore(const vector<T> &v) { // O(N)
  assert(!v.empty());
  int cnt = 0;
  T candi;
  for (auto e : v) {
    if (!cnt) candi = e;
    if (e == candi) ++cnt;
    else --cnt;
  }
  if (cnt) {
    cnt = 0;
    for (auto e : v) cnt += (e == candi);
  }
  return {cnt > v.size() / 2, candi};
}
```

7.3 Ternary Search

```
auto ternarySearch = [&](ll l, ll r) { // min
  ll lo = l, hi = r;
  while (lo + 2 < hi) {
    ll p = (lo * 2 + hi) / 3, q = (lo + hi * 2) / 3;
    if (f(p) > f(q)) lo = p; // max이면 if (f(p) < f(q))
    else hi = q;
  }
  auto res = f(lo);
  for (ll i = lo + 1; i <= hi; i++) res = min(res, f(i));
  return res;
};
```

7.4 RMQ

```
template <typename T, const T& (*op)(const T&, const T&)>
struct RMQ {
  vector<vector<T>> ac; // ac[i][j] = op(v[j:j+2^i))
  RMQ(const vector<T> &v) : ac(1, v) { // O(N logN)
    for (int i = 1, len = 1; len * 2 <= v.size(); ++i, len *=
    2) {
      ac.emplace_back(v.size() - len * 2 + 1);
      for (int j = 0; j < ac[i].size(); j++) ac[i][j] =
      op(ac[i - 1][j], ac[i - 1][j + len]);
    }
  }
  T qry(int a, int b) const { // 0-based // op[a:b]의 op()
  누적값 // O(1)
    assert(0 <= a && a <= b && b < ac[0].size());
    int i = __lg(b - a + 1);
    return op(ac[i][a], ac[i][b - (1 << i) + 1]);
  }
};
// const RMQ<int, min> rmq(v);
```

7.5 Deque Trick

```
vector<int> dequeTrickMin(int len, const vector<int> &v) { //
res[i]는 v[max(0, i - len + 1)]~v[i]의 최솟값 // 즉, 정확히
길이가 len인 구간의 최솟값은 res[len-1:n)에 저장됨
  vector<int> res(v.size());
  deque<int> dq;
  for (int i = 0; i < v.size(); i++) {
    while (!dq.empty() && v[dq.back()] > v[i]) dq.pop_back();
    // min
    // while (!dq.empty() && v[dq.back()] < v[i])
    dq.pop_back(); // max
    dq.push_back(i);
    if (i - dq.front() + 1 > len) dq.pop_front();
    res[i] = v[dq.front()];
  }
  return res;
}
// cmp 복잡할때
deque<int> dq;
auto cmp = [&](int front, int back) { // front에 구하려는 값이
들어가야 함
  // ex) min이면 return v[front] <= v[back];
};
for (int i = 0; i < v.size(); i++) {
  while (!dq.empty() && !cmp(dq.back(), i)) dq.pop_back();
  dq.push_back(i);
  if (i - dq.front() + 1 > len) dq.pop_front();
}
```

7.6 Random

```
mt19937 rng;
shuffle(v.begin(), v.end(), rng); // O(N)
uniform_int_distribution<int> dist(a, b); // a<=x<=b
uniform_real_distribution<double> dist(a, b); // a<=x<b
x = dist(rng);
```

7.7 Time

```
clock_t sttime = clock();
while(double(clock() - sttime) / CLOCKS_PER_SEC < 0.95) {}
```

7.8 Debug

```
g++ -std=c++17 -O0 -g -fsanitize=undefined
-fno-omit-frame-pointer main.cpp -o main
# -fsanitize=address
```